4 NUTRITION

7 NUTRITION PROTOCOLS

1. 90 Percent or More of the Food You Eat Should Be Unprocessed or Minimally Processed
2. Stay Hydrated
3. Eat High CP:C Ratio Protein at Each Full Meal
4. Choose Fibrous Vegetables and (Some) Fruit as Your Major Sources of Carbohydrates
5. Consume Healthy Fats as Part of Most Meals and Snacks
6. Consume Low-Sugar Fermented Foods
7. Consume Fatty Ocean Fish and/or Consider Omega-3 Supplements

Unofficial non-profit edition

Nutrition advice is everywhere, and it's often conflicting. Eat meat. Don't eat meat. Eat breakfast. Fast until noon. Carbs are the enemy. Carbs help you burn fat. Fats are key. Fats are bad. Sugar is poison. Calories, not sugar, are what matters. Seed oils are good for you. Seed oils are bad for you… It's enough to make anyone dizzy.

And there are avid debates. While some get into nuanced biochemistry, others are surprisingly basic. For example, people still argue about whether the types of foods you eat are more important than how much you eat or whether protein guidelines are off by a factor of two. People also debate whether the timing of when you eat different types of food affects body composition—that is, your body fat to lean mass ratio. Worse, dietary fat and body fat share the same name: fat. That has millions of people avoiding certain dietary fats that, were they to eat them, would likely improve their health.

I'm old enough to remember the "low fat" trend of the early 1990s—when seemingly everything was labeled low-fat or nonfat, including things that never had fat in the first place, like soda and candy, but for which labeling as "no fat!" gave the false impression they would promote body fat loss.

I also remember the 40-30-30 carbohydrate-protein-fat ratio trend, too, which was good in principle, but difficult to adhere to because—at least in my opinion—it wasn't

restrictive enough to make it easy to follow. That's right, it was too permissive to allow for easy decision-making about food choices. So people ate too much. They needed to be told what not to eat.

The point being: A clever way to start a diet trend is to demonize one type of food.

Which may partially explain why low-carbohydrate and ketogenic diets became popular in the early 2010s. Eating mostly just fat and protein requires some "no thank you" discipline but is otherwise pretty simple… until a few weeks in, when a person realizes that even a bowl of salad is off limits and that packaged "keto" snacks often contain a lot of calories… and that calories still matter.

The rise and somewhat-fall of keto seemed to usher in time-restricted eating, or intermittent fasting, which emphasizes extended periods each day (and night) with no food intake. This approach was attractive for people who find it hard to have "just a few bites" of a certain food yet find solace knowing they can eat that same food later, once they are in their "eating window." But this diet can challenge even the most disciplined, because while it has practical, psychological, and some health benefits, it doesn't work if people eat with abandon during that window.[1]

[1] Ming-Li Sun et al., "Intermittent Fasting and Health Outcomes: An Umbrella Review of Systematic Reviews and Meta-Analyses of Randomised Controlled

Oddly enough, a reptile from South America entered the picture and settled one major debate. Yes, a reptile.

Long story, made short:

The Gila monster doesn't need to eat very often. Just three to four times a year. A year!

A curious biologist asked why.

He discovered Gilas make a peptide called exendin-4 that potently suppresses appetite.

It was then discovered that humans make a peptide similar to exendin-4, called glucagon-like peptide (GLP-1).

Humans made drugs that mimic GLP-1, but a thousandfold.

People don't want to eat as much and…

They lose weight.

And a trillion-dollar weight loss drug industry was born.

More importantly for this chapter, GLP-1s really did settle that major debate about why people gain, maintain, or lose weight. They proved that sure, you can argue about which foods may indirectly impact satiety, or which foods are healthiest, but when it comes to body weight, how many calories we eat relative to how many we metabolize matters.

Trials," EClinicalMedicine (Oxford) 70 (April 2024): 102519, https://doi.org/10.1016/j.eclinm.2024.102519.

And yet there is a lot more to nutrition. A LOT more. For health. For brain health too. And not just neuron health, but also for mental health. Even for our focus and ability to learn.

There's a lot more to nutrition, in part because there are a lot of experts with different perspectives. True experts, who have done the research, observed what works or fails in the real world with women, men, kids, healthy and sick people, and with those who exercise and those who don't, or just won't.

I have had a closeup view of just how many perspectives are out there through my podcast. I've talked for hours with "pro-plant-based" scientists like Christopher Gardner, PhD, professor of medicine at Stanford University.[2] I've learned from nutritional scientist Layne Norton, PhD, and nutrition educator Alan Aragon, MS, who draw from both their academic training and their experience working with thousands of people.[3] I hosted osteopathic physician Gabrielle Lyon, DO, to discuss protein needs for both women and men, and endocrinologist Robert Lustig, MD,

[2] Andrew Huberman, "How Different Diets Impact Your Health | Dr. Christopher Gardner," Huberman Lab (podcast audio), January 29, 2024, https://www.hubermanlab.com/episode/how-different-diets-impact-your-health-dr-christopher-gardner.

[3] Andrew Huberman, "Dr Layne Norton: The Science of Eating for Health, Fat Loss, and Lean Muscle," Huberman Lab (podcast audio), November 7, 2022, https://www.hubermanlab.com/episode/dr-layne-norton-the-science-of-eating-for-health-fat-loss-and-lean-muscle; Andrew Huberman, "Dr. Layne Norton:

professor at University of California, San Francisco, about sugar.[4] (P.S. Robert is not a fan of sugar!) Justin Sonnenburg, PhD, a microbiologist at Stanford and perhaps the world expert on the human gut microbiome, spoke with me about how to support a healthy gut with specific foods.[5] Harvard Medical School psychiatrist Chris Palmer, MD, explained the now-demonstrated benefits of the ketogenic diet for depression and other mental health issues.[6] And Satchin Panda, PhD, who directs the Regulatory Biology Laboratory at the Salk Institute for Biological Studies, in San Diego, shared his extensive knowledge on the science of intermittent fasting.[7]

Tools for Nutrition & Fitness," Huberman Lab (podcast audio), August 12, 2024, https://www.hubermanlab.com/episode/dr-layne-norton-tools-for-nutrition-fitness; Andrew Huberman, "Alan Aragon: How to Lose Fat and Gain Muscle with Nutrition," Huberman Lab (podcast audio), July 7, 2025, https://www.hubermanlab.com/episode/how-to-lose-fat-gain-muscle-with-nutrition-alan-aragon.

4 Andrew Huberman, "Dr. Gabrielle Lyon: How to Exercise & Eat for Optimal Health & Longevity," Huberman Lab (podcast audio), June 24, 2024, https://www.hubermanlab.com/episode/dr-gabrielle-lyon-how-to-exercise-eat-for-optimal-health-longevity; Andrew Huberman, "Dr. Robert Lustig: How Sugar & Processed Foods Impact Your Health," Huberman Lab (podcast audio), December 18, 2023, https://www.hubermanlab.com/episode/dr-robert-lustig-how-sugar-processed-foods-impact-your-health.

5 Andrew Huberman, "Dr. Justin Sonnenburg: How to Build, Maintain & Repair Gut Health," Huberman Lab (podcast audio), October 2, 2023, https://www.hubermanlab.com/episode/dr-justin-sonnenburg-how-to-build-maintain-and-repair-gut-health.

6 Andrew Huberman, "Dr. Chris Palmer: Diet & Nutrition for Mental Health," Huberman Lab (podcast audio), November 21, 2022, https://www.hubermanlab.com/episode/dr-chris-palmer-diet-nutrition-for-mental-health.

7 Andrew Huberman, "Satchin Panda: Intermittent Fasting to Improve Health, Cognition and Longevity," Huberman Lab (podcast audio), March 13, 2023, htt

These experts disagree—sometimes vehemently—on many things… no surprise there. But they also agree about many things, and one foundational understanding in particular: that when it comes to body weight, how many calories we eat relative to how many we metabolize does matter.

In this chapter, you'll learn the consensus from nutrition experts that you likely have not heard anywhere else. I'll also share my own experience and observations, having tried many different approaches to nutrition. And I share one that is seldom discussed but that I, and many others, have used to great success. It builds on this consensus. It cuts through the noise. It puts you in the driver's seat so you can make satisfying food choices that don't sacrifice taste; that support your body composition goals; and that help ensure you get the macro- and micronutrients you need for brain and bodily health. And it can be done on a budget.

The Clean Omnivore Diet

I call the approach I'm suggesting here the "Clean Omnivore Diet." As you'll see in the pages that follow, this approach to nutrition is based in science. We begin, as we have for each of the topics in this book, with data. I provide an overview of nutrition to get everyone on the same page, and to help you navigate what follows. I do hope you'll read it, because I

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ps://www.hubermanlab.com/episode/dr-satchin-panda-intermittent-fasting-to-improve-health-cognition-and-longevity.

know it will empower you, but I'll understand if you'd like to jump to the protocols so you can learn what to eat and when for maximal health.

Body composition affects health and longevity. People with a higher ratio of lean mass to body fat tend to live longer than those with relatively higher percentages of body fat.[8]

More than 70 percent of Americans are overweight and more than a third are obese,[9] which basically means the majority of Americans are likely shortening their lifespan with their eating habits. The harshest messaging of this I've ever heard was from an MD colleague of mine who said, "You just don't meet many obese eighty-year-olds." Fill in the blanks on that.

Physicians are known for gallows humor, but this point matches the data. Body composition matters.

The advice for most people, given the above, might simply be "lose weight." But that is short sighted. I said body composition, not weight loss. If you lose weight and much of

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[8] Preethi Srikanthan and Arun S. Karlamangla, "Muscle Mass Index as a Predictor of Longevity in Older Adults," American Journal of Medicine 127, no. 6 (2014): 547–53, https://doi.org/10.1016/j.amjmed.2014.02.007; Dong Hoon Lee and Edward L. Giovannucci, "Body Composition and Mortality in the General Population: A Review of Epidemiologic Studies," Experimental Biology and Medicine 243 (2018): 1273–83, https://doi.org/10.1177/1535370218818161.

[9] National Center for Health Statistics, "Obesity and Overweight," Centers for Disease Control and Prevention (CDC), last reviewed March 6, 2026, https://www.cdc.gov/nchs/fastats/obesity-overweight.htm.

it is muscle mass, your metabolism drops, and then it's very easy to gain the weight back. (Muscle, if you recall from chapter 2, is very metabolically active, and plays a role in blood sugar management and therefore affects insulin sensitivity, which refers to how well your body responds to increases in blood sugar.) You need to maintain and/or increase muscle mass, along with bone mass, while you lose mostly body fat. And while bone is not as sexy sounding as muscle, having adequate bone mass and density is vitally important for your health and longevity too.[10]

The not-so-secret secret solution? Do resistance training, and eat at what we call maintenance calories, or slightly submaintenance calories (300–500 calories less than you need per day—more on that in a moment)… unless of course you are among those trying to gain weight, in which case you want to eat a bit more (300–500) each day. In the short run, this works. But in the long run, everyone—not just those seeking to lose or gain weight—needs to eat for health and longevity, but also for taste, and in a way that is compatible with their social life, schedule, and budget. That's a tall order.

The great news is that there's considerable overlap among diets that do just that—and that includes the Clean Omnivore

[10] C. Johansson et al., "Bone Mineral Density Is a Predictor of Survival," Calcified Tissue International 63, no. 3 (1998): 190–96, https://doi.org/10.1007/s002239900513.

Diet. They won't leave you avoiding restaurants, either. But they require understanding what food and nutrition are really about.

WHAT TO EAT

There are three macronutrients: proteins, carbohydrates, and fats. Each of these can be subdivided into categories according to how they are used by the body. Let's take a closer look at each, one at a time.

How to Think About Protein

Dietary proteins are foods rich in amino acids, compounds that our body needs to make its own proteins. All the cells and tissues of our body—hair, muscle, bone, et cetera—contain those proteins. So we need amino acids from food to keep our cells and tissues functional. Of the twenty amino acids found in the human body, there are nine that we cannot make on our own; they must come directly from outside sources—namely from dietary protein.[11] These nine amino acids are called the essential amino acids, and foods containing them are referred to as complete proteins. Many foods sourced from animals, such as red meat, poultry, fish, eggs, and shellfish, such as scallops and oysters, as well as many dairy products, contain all nine essential amino acids,

[11] Jay R. Hoffman and Michael J. Falvo, "Protein—Which Is Best?," Journal of Sports Science & Medicine 3, no. 3 (2004): 118–30, https://pubmed.ncbi.nlm.nih.gov/24482589/.

and thus are considered complete proteins. Some plant-based foods such as quinoa, hemp seeds, and soy also contain all nine essential amino acids.

Most plant-based foods, however, do not have all nine essential amino acids and must be combined with each other to be considered complete proteins. Examples include beans with rice, or nuts with lentils. There is a lot of debate (of course) as to whether animal-based proteins are superior to plant-based proteins, but the debate is often misleading because it neglects a key feature of any protein. And that is what I will refer to as the complete-protein-to-calorie ratio (CP:C ratio). Even when comparing animal-based complete proteins to one another, it is important to consider their relative CP:C ratios. I'll illustrate this concept by comparing some examples of different foods: 150 calories of chicken versus 150 calories of lean beef versus 150 calories of whole eggs versus 150 calories of tofu versus 150 calories of beans and rice. All are considered complete proteins.

Now, importantly, while you would need to eat beans and rice (or nearly any legume and any grain combination) to get all nine essential amino acids, these do not need to be eaten in the same meal to be utilized effectively. As long as you get sufficient essential amino acids across your meals over the course of a day, it works.

- 150 calories of chicken has about 28 grams of complete protein
- 150 calories of lean beef has about 25 grams of complete protein
- 150 calories of whole eggs contains about 13 grams of complete protein
- 150 calories of tofu has about 15 grams of complete protein
- 150 calories of beans and rice has about 6 grams of complete protein
- 150 calories of peanut butter on a slice of wheat bread has about 6.5 grams of complete protein
- And let’s throw in 150 calories of scallops, which has about 29 grams of complete protein.

The reason different foods and food combinations yield different amounts of complete protein at the same caloric level is because none of these foods are 100 percent protein. They also contain some fat, or in the case of plant foods, carbohydrates and fats.

Most animal-based foods like meat, eggs, and shellfish contain zero (or close to zero) carbohydrates, and varying amounts of fat. Dairy products contain protein and fat (unless defatted, like skim milk, for example), and dairy includes carbohydrates to varying degrees.

So we need to ask: Does a given “protein” contain all nine essential amino acids? And: What is its CP:C ratio?

In other words: How many calories of a given “protein” food do we need to eat in order to get enough of those nine essential amino acids?

The answer to that question matters for body composition, health, and the pleasure you experience from eating, because getting those nine essential amino acids is crucial for satiety. That’s because how much you eat at a meal is driven in part by your brain’s and body’s hunger for complete proteins. When you get enough complete protein, you tend to feel satisfied during, and for a long while after, the meal. When you don’t, you tend to feel hungry again soon after eating.

The nine amino acids are not the only reason we eat, but they are a big part of it. Below I’ll explain how, if we don’t get enough complete protein per calorie, we start to run into problems. Luckily, those problems are easy to fix. I’ll explain how to do that too.

I often hear: “But I eat protein! I eat beans, I eat peanut butter…” Well, neither of those are complete proteins, but if the beans are served with rice or if the peanut butter is spread on wheat bread, then they become complete proteins. Problem solved? Usually not. In fact, often it’s problem created. Remember, beans and rice together provide 6 grams of protein for every 150 calories. That means you would

need to eat more than four times as many calories of beans and rice (700 calories) to hit the same amount of protein (28 grams) found in 150 calories of chicken. Peanut butter on wheat bread is a similar story, given that in those 150 calories, you're only getting 6.5 grams of protein. That brings us back to CP:C ratio. Chicken has a high ratio; beans and rice or peanut butter on wheat bread has a low ratio.

To be fair, some animal-based foods also have a low CP:C ratio. Take most pork for instance. One hundred fifty calories of lean pork will give you about the same amount of protein as 150 calories of chicken or beef (about 25 grams). But most commercially sold pork products, such as bacon, ham, and pork chops, are not lean. Most contain a high percentage of fat, and often have sugar added to them too. One hundred fifty calories of low-sugar bacon has only about 10 grams of protein. Typical store-bought bacon with sugar added? About half that, at 5 grams. You would need to eat about 400 to 500 calories of bacon to get the protein found in 150 calories of chicken.

The takeaway here is that we need to prioritize eating complete protein foods, and we need to consider the CP:C ratio when selecting which to eat. While protein is not the whole story, the best nutrition plan typically places proteins with a high CP:C ratio at the center of the plate.

WHAT PROTEINS ARE BEST?

If you eat animal products, I suggest you prioritize lean cuts of beef, elk, venison, or bison, as well as fish and chicken. They have a high CP:C ratio. Many dairy products do, too, such as Greek and Bulgarian yogurt, and cottage cheese. Eggs are great as well, even though they do not technically meet the benchmark, given their ratio of protein to fat. That said, I still incorporate eggs into my diet most days. After years of public debate on whether eggs are healthy or unhealthy, the FDA consensus as of 2025 is that, yes, eggs are healthy. They appear on a list right next to salmon, 100 percent olive oil, fruits, and vegetables.[12] Our grandparents were right (again!).

Powdered proteins such as whey protein have a very high CP:C ratio and are a convenient way to get enough protein for perhaps a meal or snack per day, maybe two if you're on the go. But they should not be your only complete protein source because they lack a number of important vitamins and minerals and other micronutrients found in whole foods. You want to prioritize whole foods for that.

Plant-based sources of proteins, by contrast, tend to have a low CP:C ratio. So much so that it is best not to think of most plant-based foods as proteins at all. It's more accurate

[12] "FDA's Updated 'Healthy' Claim Definition—Factsheet," U.S. Food and Drug Administration, December 2024, https://www.fda.gov/media/184535/download?attachment.

to think of grains, like rice and quinoa, and legumes, like beans, simply as carbohydrates, because that's mainly what they are. And think of the lipid-rich plant-based sources of proteins, like nuts, as fats, because that's mainly what they are. There are exceptions, but by and large the CP:C ratio of these foods is just too low to consider them otherwise. Unless you're burning significant calories through long-duration exercise or physical labor, relying on low CP:C ratio foods to try to get enough complete protein often leads to eating excess calories and putting on body fat. And for those trying to gain weight—yes, such people are out there—devote your additional calories to high CP:C ratio proteins, carbohydrates, and healthy fats.

And before you say "He's an anti-vegan, ketogenic carnivore!" I assure you that is not the case. None of this is leading up to me suggesting a purely animal-based diet. I think vegetables and fruits are fantastic. I eat them daily and some grains too. Vegans who are willing to put in the work to seek out or prepare dishes with a variety of ingredients—such as beans and other legumes, rice and other grains, and tofu and tempeh—can get adequate amounts of the nine essential amino acids across the day without eating animal products. They just need to be mindful not to eat too many calories relative to their overall energy needs. I have seen a vegan diet work great for some people, but admittedly they tend to be very diligent about the sourcing and

combining of their plant-based foods to achieve complete proteins. Or they do a lot of cardiovascular exercise—they run marathons, triathlons, and distance races of that sort—which comes with very high caloric needs compared to that of most people.

One more consideration as you select a protein source is the bioavailability of that protein source—how much protein you can absorb as you digest it, and then how much your cells can make use of it. There is ongoing debate about this amount. Classic studies concluded that we can only absorb about 30 grams of protein per meal.[13] That amount became a sort of "law" of nutrition, especially in resistance training fitness circles that prioritize muscle protein synthesis—a process that is stimulated by weight lifting, but that is also involved in muscle maintenance for everyone, lifters and non-lifters alike. Later studies showed, however, that our bodies can use pretty much any amount of protein we ingest.[14] What changed? Mostly the conversation around protein absorption.

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[13] T. Brock Symons et al., "A Moderate Serving of High-Quality Protein Maximally Stimulates Skeletal Muscle Protein Synthesis in Young and Elderly Subjects," Journal of the American Dietetic Association 109, no. 9 (2009): 1582–86, https://doi.org/10.1016/j.jada.2009.06.369.

[14] Jorn Trommelen et al., "The Anabolic Response to Protein Ingestion During Recovery from Exercise Has No Upper Limit in Magnitude and Duration In Vivo in Humans," Cell Reports Medicine 4, no. 12 (2023): 101324, https://doi.org/10.1016/j.xcrm.2023.101324.

A portion of the protein that we consume at any given meal is used by our muscles. But there's a limit to the "dose" of protein our muscles can synthesize; that dose is around 30 to 50 grams at one time.[15] Whatever amount we consume beyond that will be absorbed by the gut and the liver, and used for things like neurotransmitter production, hair and collagen growth, immune system function, and much more. (Note that for kids still growing, and for those who resistance train regularly, the amount directed for protein synthesis is probably closer to 40 to 50 grams per meal, with the excess available for other tasks.)

Does this mean that you must consume 30 to 50 grams of protein at every meal if you want to maintain your current muscle mass? Nope. Research on intermittent fasting (a good setting to investigate this question because daily calories are consumed in a small window rather than as three meals spread across a day) found that lean muscle mass is preserved as long as enough total protein is consumed in a day.[16]

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[15] Brad Jon Schoenfeld and Alan Albert Aragon, "How Much Protein Can the Body Use in a Single Meal for Muscle-Building? Implications for Daily Protein Distribution," Journal of the International Society of Sports Nutrition 15, no. 10 (February 2018), https://doi.org/10.1186/s12970-018-0215-1.

[16] Damoon Ashtary-Larky et al., "Effects of Intermittent Fasting Combined with Resistance Training on Body Composition: A Systematic Review and Meta-Analysis," Physiology & Behavior 237 (August 2021): 113453, https://doi.org/10.1016/j.physbeh.2021.113453.

All of this explains why people versed in nutrition don't just talk about how much protein we can absorb, but the metabolic fate of the proteins we eat. In fact, there are times when assimilating 100 grams (!) per meal is possible.[17] Does that mean you should eat 100 grams of protein per meal? Not necessarily. First off, how many meals are you eating? And how much protein do you need? Again, we return to the data.

HOW MUCH PROTEIN DO YOU NEED?

Most people should aim to consume 1 gram of protein, ideally from high CP:C ratio protein sources, per pound of body weight each day. It's not crucial that you eat exactly that much every day. I suggest it as a target from which you experiment a bit to find your protein needs. If you find yourself too full or otherwise unable to hit it, that's okay. You should never feel like you're force-feeding yourself just to hit a protein goal. Strive to hit the 1 gram of protein per pound of body weight (or close to it) most days. And if you prefer to divide your protein into multiple meals and snacks per day, do that. If, however, you prefer to eat twice a day and ingest more protein at those meals, do that, even if you're resistance training specifically to gain muscle. While it's true that spreading out your protein across four to five meals and snacks per day can provide a marginal increase in muscle protein synthesis if you're resistance training and

[17] Trommelen et al., "Anabolic Response to Protein Ingestion."

aiming to gain the maximum amount of new muscle, the difference it makes won't be meaningful for the majority of people. So figure out the timing of your meals and snacks that works best for your hunger, energy levels, sleep, and schedule. Be mindful of, but don't obsess about, that 1 gram of protein per pound of body weight target.

Some people worry that eating too much protein can cause kidney problems or gout, but the data show that only becomes an issue if people consume something in the vicinity of more than 3 grams of protein per pound of body weight—an absurdly high amount that most people would struggle to reach.[18] So don't fear eating high CP:C ratio foods. That's one more debate that's easy to settle.

Another debate: There is also talk that deliberately eating far less protein than we need can help us live longer. The theory is that eating protein stimulates a rise in the level of mTOR (which stands for mammalian target of rapamycin), a protein related to cell growth that, when suppressed, has been shown to extend lifespan in animals.[19] So far, however, the data point to "NOPE" on whether a

[18] Ioannis Delimaris, "Adverse Effects Associated with Protein Intake Above the Recommended Dietary Allowance for Adults," ISRN Nutrition 2013 (July 2013): 126929, https://doi.org/10.5402/2013/126929.

[19] Michelle M. Farnfield et al., "Why Protein Ingestion Activates mTOR-dependent Signalling After Resistance Exercise in Young Men: A Double-Blinded Randomized Controlled Trial," Nutrients 1, no. 2 (December 2009): 263–75, https://doi.org/10.3390/nu1020263; Grace Y. Liu and David M. Sabatini, "mTOR at the Nexus of Nutrition, Growth, Ageing and Disease,"

low-protein-to-suppress-mTOR diet helps humans live longer.[20]

Entire books could be devoted to any of the topics above but for now, let's just keep the CP:C ratio in the back of our minds as an important metric to consider when making food choices. Prioritizing eating enough foods with a high CP:C ratio to reach that 1 gram per pound of body weight removes a lot of confusion in the "what to eat" debate.

How to Think About Carbohydrates

Carbohydrates, or carbs, as we often call them, have become appropriately controversial in the last few decades. I say that because carbs include both healthy and very unhealthy varieties. And no, that's not determined by just whether a carb is complex versus simple, a term that refers to their molecular structure (more on that later). Some carbs quell hunger. Others can make you eat with abandon. Sometimes it depends what those carbs are combined with. Carbs combined with fat and with sugar, for example, can make people with a sweet tooth feel like an addict who just scored: focused on getting more. It becomes very easy to overeat.

Nature Reviews Molecular Cell Biology 21 (January 2020): 183–203, https://doi.org/10.1038/s41580-019-0199-y; Joan B. Mannick and Dudley W. Lamming, "Targeting the Biology of Aging with mTOR Inhibitors," Nature Aging 3 (May 2023): 642–60, https://doi.org/10.1038/s43587-023-00416-y.

[20] Liu and Sabatini, "mTOR."

Should you just cut out all carbs? Well, cutting out carbs completely will make you lose weight fast, but mostly you lose water weight as carbs cause you to "hold" water.[21] For most people, I don't suggest suddenly cutting out carbs completely, though cutting back on certain types of carbs can be beneficial for many people. I've seen folks with autoimmune issues resolve them by moving to a carnivore (meat, salt only) or ketogenic (very low carb) diet. Then, most gradually reintroduce carbs over time, learning which carbs are problematic for them and which are not. I have also seen people who could not lose body fat do so on very low carb diets. Ketogenic diets have also been shown to be an effective treatment for some forms of drug-resistant epilepsy[22] and in some cases for schizophrenia[23] and depression.[24] Also, some people just "feel better," both mentally and physically, on a zero or near-zero carbohydrate diet. Incidentally, I am not one of those people. I love fruits

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[21] Robert Oh et al., "Low-Carbohydrate Diet," StatPearls (internet) (Treasure Island, FL: StatPearls Publishing, January 2026).

[22] Kirsty J. Martin-McGill et al., "Ketogenic Diets for Drug-Resistant Epilepsy," Cochrane Database of Systematic Reviews 6, article no. CD001903 (2020), https://doi.org/10.1002/14651858.CD001903.pub5.

[23] Shebani Sethi et al., "Ketogenic Diet Intervention on Metabolic and Psychiatric Health in Bipolar and Schizophrenia: A Pilot Trial," Psychiatry Research 335, article no. 115866 (2024), https://doi.org/10.1016/j.psychres.2024.115866.

[24] Drew D. Decker et al., "A Pilot Study Examining a Ketogenic Diet as an Adjunct Therapy in College Students with Major Depressive Disorder," Translational Psychiatry 15, article no. 322 (2025), https://doi.org/10.1038/s41398-025-03544-8.

and vegetables and I eat starches, but I am thoughtful about which and, to some extent, when.

I also know people who have done well on an "animal-based" diet, which recommends people eat only meat, fish, eggs, and dairy. It's actually a misnomer because the actual animal-based diet made popular by Paul Saladino, MD, includes not just meat, eggs, fish, and dairy, but also fruits such as dates, bananas, cucumbers, and tomatoes (yes, those last two are fruits).[25] This diet is much easier to adhere to, and arguably much healthier in the long run than a pure-meat carnivore diet. The controversial part is that it is ostensibly "anti-vegetables." In the last year, I have seen public messaging from Paul that asparagus now gets the green light. (No pun intended.)

The low starch, minimal-to-zero vegetables thing can be a real challenge. The vast majority of us are—and want to continue to be—omnivores, meaning we would like to eat meat but also fruits, vegetables, and, yes, other carbohydrates. So with that in mind, let's parse this category of macronutrients.

First, there is no such thing as an essential carbohydrate—you can survive by eating only proteins and fats. But all our cells run on glucose and, assuming we eat

[25] You can read more about this diet at Paul Saladino, MD, www.paulsaladinomd.com.

them, carbs readily get converted to glucose. That glucose is absorbed into our bloodstream—which leads to a rise in blood sugar, which in turn triggers the release of hormones like insulin that help manage this fuel. Typically, the breakdown of carbs into glucose is what our bodies prioritize for energy, then fats, and finally protein. That's an oversimplification, but all things being equal, that's the sequence.

To fully understand how carbs work, you need to know a bit more about how carbs are processed alone and in combination with other foods, and how they impact appetite. Carbohydrates can be broadly divided into complex carbohydrates and simple carbohydrates. Complex carbohydrates can be further (and still broadly) divided into starchy complex carbs and fibrous complex carbs. As a general rule, starchy complex carbohydrates, aka starches, are complex carbohydrates that completely or mostly dissolve in your mouth (after they are cooked or manufactured). Examples include rice, potatoes, beans, yams, pasta, lentils, bread, chips and crackers, and so on. When consumed with little or no added fats or sugars, non-processed starchy carbs are what some call clean starches. So if you're wondering—chips and crackers, while mostly starch, are not clean. Although fairly calorie dense, clean starches like oatmeal, potatoes, rice, beans, and lentils are terrific sources of energy because they are more slowly

converted to glucose by the body. Breads made without excess sugar or preservatives such as rye bread, sourdough, and unbleached wheat bread can also be considered clean(ish) starches.

That being said, you can overconsume starchy carbs.[26] Four things tend to lead you there:

- If you don't eat enough high CP:C ratio protein along with them.
- If you add sugar to them including jams, honey, syrup, and more.
- If you add fat to them like butter, oil, or cheese.
- If you add both sugars and fats.

Starchy carbs are easy to overconsume, in no small part because they literally melt in your mouth. They are flavorful especially when combined with fat and salt (think bread alone versus bread and butter with salt or bread dipped in

[26] A wealth of peer-reviewed literature documents the problems with how starch is consumed in modern Western diets. Here are a few examples for further reading: Erik E. J. G. Aller et al., "Starches, Sugars and Obesity," Nutrients 3, no. 3 (2011): 341–69, https://doi.org/10.3390/nu3030341; Tera L. Fazzino et al., "Meal Composition During an Ad Libitum Buffet Meal and Longitudinal Predictions of Weight and Percent Body Fat Change: The Role of Hyper-Palatable, Energy Dense, and Ultra-Processed Foods," Appetite 167, article no. 105592 (December 1, 2021), https://doi.org/10.1016/j.appet.2021.105592; Tera L. Fazzino et al., "Hyper-Palatable Foods: Development of a Quantitative Definition and Application to the US Food System Database," Obesity 27, no. 11 (2019): 1761–68, https://doi.org/10.1002/oby.22639.

olive oil with salt, or french fries, or macaroni and cheese). And when starches are combined with fat and sugar and just enough salt to bring out the sweet taste (think cheesecake, doughnuts, chocolate cake, pie, ice cream, and most breakfast cereals nowadays) most people find them super tasty, and hard to not overeat. Processed starchy carbs, like chips, crackers, store-bought cookies, frozen pizza, burritos, et cetera, are basically starches with fats and/or sugar blended evenly throughout, so every bite tastes identical—identically scrumptious that is. Chips and crackers are especially easy to overconsume because they activate the “eat more of this!” circuit in your brain and body, without any of the usual satiety promoting features of whole foods. They tend to have a very low CP:C ratio, and they are incredibly calorie dense for their size. You can easily eat 800 calories of chips without feeling full… and in many cases doing that will tend to increase your appetite. By comparison, you would have to eat about 16 ounces (a pound!) of cooked steak to get to the same 800 calories. I’m betting by the tenth ounce of steak, you’re going to feel very satiated, and would stay that way for many hours.

I don’t want to demonize starches. I eat them. I love them. Rice and pasta dishes and a great thick-crusted rye, or sourdough bread with olive oil or butter and salt are some of my favorite foods. I also think most people can enjoy a slice of cake or pie, a scoop of ice cream, a hot dog, a slice or two

of pizza, and other carb-fat-sugar combinations now and again without any issue. But even if they are homemade, or from a bakery or restaurant that uses quality ingredients, they should not be a staple of anyone's diet. And the packaged, processed versions are certainly damaging to our health. This is true for a variety of reasons, but their low micronutrient value and their tendency to make people overeat are just two.[27]

SO WHICH CARBS SHOULD WE EAT?

Most people, unless following a ketogenic, carnivore, or animal-based diet, should consume some clean carbs one to three times per day. How much? A simple solution is to scale your starchy carb intake with the amount and type of physical activity you're doing. Unless you are still growing, are very lean (less than 10 percent body fat for a man and less than 15 percent for a woman), or are doing a lot of cardio (1.5 hours or more per day on average) or hard manual labor (3-plus days a week), it is probably best to limit your starch intake to a third or less of your daily calories. Strive to make that from clean starches, and don't

[27] Carlos A. Monteiro et al., "Ultra-Processed Foods: What They Are and How to Identify Them," Public Health Nutrition 22, no. 5 (2019): 936–41, https://doi.org/10.1017/S1368980018003762; Kevin D. Hall et al., "Ultra-Processed Diets Cause Excess Calorie Intake and Weight Gain: An Inpatient Randomized Controlled Trial of Ad Libitum Food Intake," Cell Metabolism 30, no. 1 (July 2019): 67–77.e3, https://doi.org/10.1016/j.cmet.2019.05.008.

add more than one to two tablespoons of fat (butter or olive oil, etc.) to any one starch serving. And don't add sugar.

If you do intense cardio or hard resistance training three or more times a week (meaning you are doing compound multi-joint exercises and take your work sets to failure; see chapter 2 for details), then consider also consuming some clean starches in the meals that follow your training session. Ideally, eat some fruit too. Long, intense cardio and resistance training of the sort described in chapter 2 can deplete your glycogen reserves.[28] You would be wise to consume 30 to 75 grams of high CP:C ratio protein in that same meal. By the way, unless it's very intense, a yoga class or bodyweight workout won't meaningfully deplete your glycogen reserves. Nor will half an hour of Zone 2 running or easy laps in the swimming pool. That is not what I'm talking about. But if you push yourself running, swimming, or cycling or training with heavy (for you) weights for forty-five to seventy-five minutes, or do a truly hard Pilates workout, then you likely need to replenish your glycogen reserves. Eating some clean starches (and fruit and some high CP:C ratio protein) is a great way to do that. In fact, unless you're in ketosis and familiar with that state, failure to replenish your glycogen, or dropping your starches to zero

[28] Bob Murray and Christine Rosenbloom, "Fundamentals of Glycogen Metabolism for Coaches and Athletes," Nutrition Reviews 76, no. 4 (2018): 243–59, https://doi.org/10.1093/nutrit/nuy001.

for one to two days in a row, can lead to sleep challenges, headaches, and brain fog. Resistance training with any real intensity when you haven't eaten any starches in the preceding days can be extremely difficult. Most need those clean starches, just not in excess. And if you resistance train with intensity—taking sets to failure or close to it (as described in chapter 2)—you'll want to eat those clean starches to have the glycogen your muscles need to continue your workouts.

On that note, if you follow a ketogenic, animal-based, or carnivore diet, you'll have to make up for the lack of starch with calories from fat and/or protein, and be mindful of replenishing your glycogen in other ways after hard workouts. Protein can be converted to glucose by your liver through a process called gluconeogenesis. I know very few people who consistently follow a pure ketogenic or pure carnivore diet who also consistently resistance train with intensity, but they exist. Shawn Baker, MD, a popular public educator and proponent of the carnivore diet, is a good example of someone for whom this approach works. I've seen a fair number of low/no-carb folks who do pull-ups and push-ups and other body weight exercises as well as cardio. But for most people, if you train with weights in a manner similar to how I described in Exercise Protocol 2, you're going to want to eat starches, especially in the meal following training.

There's another type of carbohydrate to discuss: fibrous carbohydrates, which as their name suggests, are high-fiber-containing carbohydrates like broccoli, cucumbers, lettuce, kale, chard, asparagus, and so on. These tend not to dissolve in your mouth, even if they are cooked first. It's hard to overstate how great complex fibrous carbs are for us. Indeed, most of us should be trying to eat more of them. They are rich with vitamins, minerals, and other micronutrients and, as their name suggests, they contain a lot of fiber. We're told all the time that we need to eat more fiber, but trying to get fiber mainly from starches is a mistake. That tends to make people gain body fat, and I'm hearing from more and more people about allergy and gut issues due to eating too many grains. Look to get most of your fiber from vegetables. Though be mindful to chew your fibrous carbs well or they can be hard on your digestive tract. And it's good to note if any specific fibrous carbs do not agree with you (you'll know!). But as a general rule, ingesting more fiber from vegetables and from low-sugar fruits is associated with better health outcomes and longevity.[29] Fibrous carbs also slow the absorption of

[29] Nicola Veronese et al., "The Impact of Dietary Fiber Consumption on Human Health: An Umbrella Review of Evidence from 17,155,277 Individuals," Clinical Nutrition 51 (August 2025): 325–33, https://doi.org/10.1016/j.clnu.2025.06.021; Sara B. Seidelmann et al., "Dietary Carbohydrate Intake and Mortality: A Prospective Cohort Study and Meta-Analysis," Lancet Public Health 3, no. 9 (2018): E419–28, https://doi.org/10.1016/S2468-2667(18)30135-X.

glucose into your bloodstream for a steadier rise in your glucose levels, and they make you feel full.[30] Our sense of satiety comes from a combination of having eaten enough essential amino acids and fatty acids, and from our glucose levels having risen (but not too high), but also from the mechano-sensation of the stretch sensors in our gut… which is nerd-speak for "feeling full."

There are even more benefits. As I mentioned, vegetables contain remarkably high levels of micronutrients, like vitamins, minerals, and more, and deliver those with very few calories. Whether they taste "yum," "yuck," or "meh" to you tends to be a function of age (know many kids under eight that love chard?), and how these foods are prepared. It should be obvious that adding a lot of oil or butter or sugary sauces can quickly make fibrous vegetables easy to overeat, and very calorie dense—especially if the vegetables are cooked—so add those sparingly.

And we should all avoid fried vegetables, fried starches, and fried meats. Fried foods are terrible for your cardiovascular health and longevity and even have been linked to poorer

[30] Masoumeh Akhlaghi, "The Role of Dietary Fibers in Regulating Appetite, an Overview of Mechanisms and Weight Consequences," Critical Reviews in Food Science and Nutrition 64, no. 10 (2024): 3139–50, https://doi.org/10.1080/10408398.2022.2130160; Eliana Bistriche Giuntini et al., "The Effects of Soluble Dietary Fibers on Glycemic Response: An Overview and Futures Perspectives," Foods 11, no. 23 (2022): 3934, https://doi.org/10.3390/foods11233934.

mental health.[31] They are high in calories, often high in sodium, and can contain trans fat, which has been linked to heart disease and other bad outcomes. If you're serious about your health, ditch them. If you indulge occasionally with a doughnut or French fries, no big deal. But if eating fried foods is a daily, or even a weekly habit, it will likely catch up to you.

There's one last carbohydrate relevant to this conversation. Simple carbohydrates are what are often referred to as sugars—although that can lead to a lot of confusion.

Unfortunately, when most people hear the word sugar they immediately think of white table sugar, or sucrose, but simple sugars exist in many forms. Naturally occurring simple sugars in foods that have fiber and fairly high water content, like fruits or vegetables, are absorbed more slowly. The simple sugars found in honey or maple syrup, which have very little fiber, and no fat, are absorbed very quickly. Lactose, the simple sugar found in full-fat milk, is absorbed slowly because of the presence of fat. The point is that

[31] Pei Qin et al., "Fried-Food Consumption and Risk of Cardiovascular Disease and All-Cause Mortality: A Meta-Analysis of Observational Studies," Heart 107 (2021): 1567–75, https://doi.org/10.1136/heartjnl-2020-317883; Taraka V. Gadiraju et al., "Fried Food Consumption and Cardiovascular Health: A Review of Current Evidence," Nutrients 7, no. 10 (October 2015): 8424–30, https://doi.org/10.3390/nu7105404; Anli Wang et al., "High Fried Food Consumption Impacts Anxiety and Depression due to Lipid Metabolism Disturbance and Neuroinflammation," Proceedings of the National Academy of Sciences U.S.A. 120, no. 18 (2023): e2221097120, https://doi.org/10.1073/pnas.2221097120.

sugars vary in type, and how fast they are absorbed depends on the other macronutrients (proteins, fats, and carbs) you eat with them.

Maybe you've heard that "sugar is poison." This is not true of the sugar in fruit and vegetables, or in many full-fat milk products. If your goal is to have stable blood sugar and energy levels, those sources of sugar should account for about 95 percent of your sugar intake. And while you must be careful not to overconsume maple syrup or honey and other natural liquid sweeteners—they are very dense with sugar and calories—they are not poison. So adding a small bit of pure honey or pure maple syrup to your food or drinks every so often is no big deal.

Refined sugars are another story. Refined sugars are simple sugars that have been stripped of their fiber or fat content and are often chemically modified to taste extremely sweet. Some are so sweet that consuming them may change your sweet taste threshold and make otherwise savory or sweet tasting foods taste more bland.[32] Luckily that effect is reversible if you cut out refined sugars even just for a few days.

Sucrose, which is refined from sugarcane, is the most basic example of a refined sugar. A bit of it here or there, like table

[32] Carole Bartolotto, "Does Consuming Sugar and Artificial Sweeteners Change Taste Preferences?," Permanente Journal 19, no. 3 (2015): 81–84, https://doi.org/10.7812/TPP/14-229.

sugar in dessert, is generally not a problem for most people. Keeping refined sugars to 5 percent or less of your sugar intake is a good idea. Or, if you loathe tracking that sort of thing, simply limit yourself to one serving—a scoop of ice cream or piece of cake—per week. To some of you that may seem spartan. To others, that may seem indulgent. Later we will discuss sugar cravings, and how to beat them. In the meantime, I can assure you that it is possible to stop craving sugar, and cutting out refined sugars entirely, or close to it, is the first step in that process.

Alas, these days, sucrose or high-fructose corn syrup are put into all sorts of foods that we think of as savory (e.g., soups, cereals, sauces, and salad dressings) in order to make them taste sweeter than they would otherwise. So be a bit more vigilant and read food labels if your goal is to avoid added sugars of any kind.

Sugar is immensely reinforcing—ingesting it makes you want to eat more. So while people debate if sugar is truly "addicting," most everyone agrees that consuming refined sugars is highly reinforcing, and is not healthy—or at least offers little, if any, nutritional benefit.[33]

So while I don't think that sugar is "poison," nor do I feel it meets the classic definition of addictive, which would mean

[33] Di Qin et al., "Sugar Addiction: Neural Mechanisms and Health Implications," Brain and Behavior 15, no. 7 (2025): e70338, https://doi.org/10.1002/brb3.70338.

its use is progressive and destructive to multiple areas of life, I do, however, think eating too many sugars, and especially refined sugars, is bad for us. It can lead to gaining body fat due to the overconsumption of calories with little nutritional value, as well as insulin sensitivity problems that, at the extreme, can cause diabetes and other metabolic issues.[34] It can even lead to mental health challenges, such as depression.[35]

And sugar is everywhere. Consider that complex starchy carbohydrates—like wheat, rice, potatoes—if refined to remove their fiber content become more like simple carbohydrates. Foods containing refined flours, such as bread, pasta, cereals, crackers, and most packaged snack foods, are good examples. And they often contain added sugar as well. Even if the packages show some amount of fiber, these foods almost always act more like sugars in the body—they get absorbed fast, and almost always lead to an overconsumption of calories with little nutritional value.

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[34] Emily J. Endy et al., "Added Sugar Intake Is Associated with Weight Gain and Risk of Developing Obesity over 30 Years: The CARDIA Study," Nutrition, Metabolism and Cardiovascular Diseases 34, no. 2 (February 2024), https://doi.org/10.1016/j.numecd.2023.10.022; Quanhe Yang et al., "Added Sugar Intake and Cardiovascular Diseases Mortality Among US Adults," JAMA Internal Medicine 174, no. 4 (April 2014): 516–24, https://doi.org/10.1001/jamainternmed.2013.13563.

[35] Anika Knüppel et al., "Sugar Intake from Sweet Food and Beverages, Common Mental Disorder and Depression: Prospective Findings from the Whitehall II Study," Scientific Reports 7, no. 1 (2017): 6287, https://doi.org/10.1038/s41598-017-05649-7.

While eating something fibrous (like a salad) or fatty (like almond butter) with these refined complex carbs will typically slow their absorption, the outcome is almost always the same: an overconsumption of calories with little nutritional value. I'm not out to demonize the packaged food industry but their goal is simple: to get you to buy and consume more of what they sell. And it's clear they've been successful in reaching that goal given the prevalence of these foods and the declining health of those who consume them. We have plenty of evidence now linking ultra-processed foods with serious health issues, including heart disease, type 2 diabetes, obesity, depression, and impaired sleep.[36] The more cryptically problematic foods, in my opinion, are refined carbohydrates with added sugars. We don't think of them as highly processed compared to, say, chips or microwave meals, but they are nearly as bad in terms of our propensity to overeat them. Luckily there is an easier and much more satisfying way to eat. One that is affordable too. But first, we have to understand fats.

How to Think About Fats

First off, it's a real shame this category of macronutrients is called fats because that immediately makes people think about body fat, which almost no one wants. But you do want

[36] Melissa M. Lane et al., "Ultra-Processed Food Exposure and Adverse Health Outcomes: Umbrella Review of Epidemiological Meta-Analyses," BMJ 384 (February 2024): e077310, https://doi.org/10.1136/bmj-2023-077310.

fats in your diet. Lipids would have been a better choice in my opinion but alas, here we are.

Fats are lipids, and they are calorically dense: 9 calories per gram compared to 4 calories per gram for carbs or protein. And fats are vitally important. We can't live without them. In fact, a large part of each of our cells is made of fats, and thus consuming enough fats is crucial for cellular function and repair throughout the body. You need to eat fats, but the right kinds and the right amounts. So get it out of your mind that fats are bad. Remember, there are no essential carbohydrates, but you absolutely must eat protein and fat to survive.

Fats can be broadly divided into saturated fats and unsaturated fats.

The simplest way to recognize a saturated fat is that these are generally solid at room temperature. The saturated fat in cheeses and meats are good examples of this. While some non-animal sources such as coconuts and Brazil nuts contain saturated fat, most saturated fats come from meat, dairy, and eggs. People love to debate if saturated fats raise your cholesterol, and it's become a circular argument because yes, they do, but then the discussion turns to if cholesterol levels, HDL and LDL (classically referred to as the good and bad cholesterols, respectively), are truly meaningful in predicting heart disease and longevity, or if there are more important

metrics. The whole thing can get rather confusing.[37] Some people who eat low saturated fat diets have high LDL, due to genetics or other factors like age; some people who eat high saturated fat diets don't have high LDL, but they also tend to eat a lot of fiber, and not much starch and sugar. Pretty soon, talk turns to how insulin from carbs and not just fats impacts LDL… and the debate rages on.

SO HOW MUCH FAT DO I NEED IN MY DIET?

If you are eating animal-based proteins daily, and that includes eggs, red meat, or full-fat dairy, chances are good that you are getting enough saturated fat. If you are vegetarian or entirely plant-based, you might consider incorporating eggs and dairy (if vegetarian), along with coconut or coconut oil, macadamia nuts, cashews, or other foods and oils that have saturated fat.

Although debated, the data show that you want only moderate amounts of saturated fats in your diet because high levels of saturated fat consumption can increase systemic inflammation—an effect that is only made worse if you also ingest sugar.[38] Some might argue that is a consequence of

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[37] For a helpful review of the data and debates around dietary fat, see Nita G. Forouhi et al., "Dietary Fat and Cardiometabolic Health: Evidence, Controversies, and Consensus for Guidance," BMJ 361 (June 2018): k2139, https://doi.org/10.1136/bmj.k2139.

[38] Jade Berg et al., "Saturated Fatty Acid Intake Is Associated with Increased Inflammation, Conversion of Kynurenine to Tryptophan, and Delta-9 Desaturase Activity in Healthy Humans," International Journal of Tryptophan

eating saturated fats and carbs, not the fats per se. Yet, even if you're following a strictly ketogenic diet, excess saturated fat consumption is not good for your immediate or long-term health.[39] So regardless of whether you are an omnivore, keto, or vegetarian, consuming leaner cuts of meat and emphasizing proteins with high CP:C ratios will help keep your saturated fat intake relatively low. It's worth noting that I regularly do blood work to keep an eye on my cholesterol levels and other blood biomarkers. I think this is a good practice to truly understand how diet is affecting your blood lipids. Function Health, a company I'm an advisor to, makes this type of testing quite affordable. There are other companies entering the space as well and helping to drive the cost of blood tests down.

WHAT ABOUT UNSATURATED FATS?

Unsaturated fats tend to be liquid at room temperature. There are two types: monounsaturated and polyunsaturated fats, both of which come mainly from fish, seeds, olives, and avocados. (More on the benefits of unsaturated fats in Nutrition Protocol 5.)

Research 13 (January 2020): 117864692098194, https://doi.org/10.1177/1178646920981946; Xiao Ma et al., "Excessive Intake of Sugar: An Accomplice of Inflammation," Frontiers in Immunology 13, no. 13 (August 31, 2022), https://doi.org/10.3389/fimmu.2022.988481.

39 Jibran A. Wali et al., "Cardio-Metabolic Effects of High-Fat Diets and Their Underlying Mechanisms—a Narrative Review," Nutrients 12, no. 5 (May 2020): 1505, https://doi.org/10.3390/nu12051505.

And for those of you following the ongoing debate about seed oils—canola, grapeseed, corn oils, et cetera—I give my take on them in Nutrition Protocol 5. Some evidence points to the idea that if seed oils are used to replace someone's saturated fat intake, their health improves.[40] Though that overlooks the fact that consuming seed oils often tends to correlate with eating more processed foods, and carb-fat combination foods. And some argue excessive heat during seed oil processing makes them potentially hazardous, which is why this topic is contentious. (Nutrition Protocol 5 gives you a fuller picture.)

My recommendation is to save yourself a giant headache and when given the choice between cooking with seed oils or other fats, go with pure olive oil or avocado oil, or perhaps some small amount of butter, tallow, or ghee. If you need to cook at high temperatures, avocado oil is your best bet given that it has a high smoke point. Smoke point is the

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[40] Lijuan Han et al., "Environmental Stress-Stable Bigels Structured by Sitosterol–Oryzanol and Guar Gum: Dual-Network Mechanism and Food-Relevant Applications," Food Hydrocolloids 173 (April 2026): 112248, https://doi.org/10.1016/j.foodhyd.2025.112248; Dariush Mozaffarian, "Dietary and Policy Priorities for Cardiovascular Disease, Diabetes, and Obesity: A Comprehensive Review," Circulation 133, no. 2 (January 2016): 187–225, https://doi.org/10.1161/CIRCULATIONAHA.115.018585; Lukas Schwingshackl et al., "Effects of Oils and Solid Fats on Blood Lipids: A Systematic Review and Network Meta-Analysis," Journal of Lipid Research 59, no. 9 (July 2018): 1771–82, https://doi.org/10.1194/jlr.P085522; Skaeff C. Murray et al., "Dietary Induced Changes in Fatty Acid Composition of Human Plasma, Platelet, and Erythrocyte Lipids Follow a Similar Time Course," Journal of Nutrition 136, no. 3 (March 2006): 565–69, https://doi.org/10.1093/jn/136.3.565.

temperature at which an oil starts to smoke under high heat—which can lead to it degrading as the fats oxidize, and that produces harmful compounds linked to inflammation. When it comes to seed oils, I personally avoid them, opting to cook and make salad dressing from olive oil and using small amounts of butter here and there. No one debates olive oil's health benefits.[41] And it's delicious.

THE BEST DIET FOR YOU

We have covered a lot here. But if you're realizing that the use of words like proteins, carbs, sugars, and fats requires distinguishing the different types of each and how they interact, then you're on track to devise the best diet for you. The good news: There are several different diet frameworks that allow for that, so you have options. Here they are:

Low Carb/Keto

Technically speaking, as discussed, we can get all we need to survive from consuming only proteins and fats, and can skip carbohydrates entirely. Done right, the ketogenic diet tends to raise energy levels and assist weight loss, because you

[41] Marta Guasch-Ferré et al., "Consumption of Olive Oil and Risk of Total and Cause-Specific Mortality Among U.S. Adults," Journal of the American College of Cardiology 79, no. 2 (January 18, 2022): 101–12, https://doi.org/10.1016/j.jacc.2021.10.041; Pamela de Aguiar Sobral et al., "Health Outcomes Related to the Consumption of Olive Products: A Brief Review," Plant Foods for Human Nutrition 78, no. 4 (November 2023): 643–53, https://doi.org/10.1007/s11130-023-01119-w.

excrete a lot of water that starches normally cause you to retain. Ketogenic diets have also been shown to help resolve symptoms of epilepsy, as mentioned already, as well as major depression, bipolar depression, even schizophrenia, in some people, and there is growing evidence it may help for people with gut and autoimmune issues.[42] But to do it right means eating mostly fat, some protein, and very, very little carbohydrates, along with all the vitamins and minerals you need—which takes a lot of discipline and attention to detail. Most people just won't do that for long because they can't imagine saying goodbye to bread, fruit, pasta, salads, et cetera, forever. Still, it might be the right choice for you. Ask yourself this: Are you willing to give up starches and fruit?

Pure Vegan

By pure vegan, I mean no animal products whatsoever. That's possible. A vegan friend of mine (yes, I have some vegan friends) is a "platinum star" vegan, meaning he's never knowingly eaten any animal products. The real challenge with this diet is getting enough high CP:C ratio proteins without eating too many calories, and to make sure you don't overconsume processed foods, just because many

[42] Alexis A. Hung et al., "The Ketogenic Diet: An Anti-Inflammatory Treatment for Schizophrenia?," Journal of Inflammation Research 18 (December 2025): 16761–71, https://doi.org/10.2147/JIR.S540859; Carmen Rubio et al., "Impact of the Ketogenic Diet on Neurological Diseases: A Review," Life 15, no. 1 (2025): 71, https://doi.org/10.3390/life15010071.

of them—such as French fries or coconut ice cream—are vegan (and convenient and delicious). I know people who have been vegans for many years and thrive. One thing all vegans must be careful of is muscle loss, especially as they age. Resistance training can help offset that, but getting those high CP:C ratio proteins is key, of course, too.

Animal-Based

This diet has a very misleading name, as, yes, it includes meat, eggs, and fish, but also fruit, honey, dates, tomatoes, avocados, coconut, hard cheeses, and other full-fat dairy. Sounds decadent, no? It is. But it also takes some discipline to follow because it's an omnivorous diet minus any starches or vegetables. That means no rice, no oatmeal, no bread, no broccoli, no lettuce, and on and on. Yes, it intentionally lacks vegetables. And the only processed food you'll eat on this diet is hard cheese like Parmesan. This diet is possible for pescatarians who want to eat fatty ocean fish such as salmon and sardines. It's also fine for people who don't like red meat but eat poultry. I have heard from many people who have achieved impressive body composition results and improved their health outcomes by following an animal-based diet. That said, because it includes a lot of calorie-dense foods, you still need to watch your total caloric and saturated fat intake. In my view, any dramatic effectiveness of this diet is likely to be multifactorial,

including that it limits starch-fat and starch-sugar combinations, and that it makes no room for processed foods or refined sugars of any kind. It tends to quell hunger fairly well and contribute to stable blood sugar. That said, it can tend to lead to very high LDL levels in some individuals, which can be correlated with increased risk of heart disease and stroke, a result of fatty plaque building up in arteries.[43] If you opt to follow an animal based diet, I suggest getting a calcium CT scan of your heart before starting and again a year later to see if it causes you to add plaque to your arteries. I know people for whom it has made plaque worsen, and those for whom (independent of their LDL levels) it has not caused increased blockages. We still have a lot to learn about the relationship between LDL, markers like ApoB, diet, and cardiovascular diseases. Meanwhile, assess your current cardiovascular risk profile and how your diet—animal-based or otherwise—is impacting it.

Clean Omnivore

My diet of choice is what I call the Clean Omnivore Diet because it checks off all, not just a few, of the key pillars of effective, health-promoting, delicious, cost-effective, and practical nutrition, which makes it easy to stick to. If we take

[43] Joyce A. Nettleton et al., "Saturated Fat Consumption and Risk of Coronary Heart Disease and Ischemic Stroke: A Science Update," Annals of Nutrition & Metabolism 70, no. 1 (April 1, 2017): 26–33, https://doi.org/10.1159/000455681.

a step back from the debates, and ask what we need and want from a great overall diet, the answer that emerges is this:

- Plenty of protein from high CP:C ratio sources: Aim for 1 gram per pound of body weight of high CP:C ratio proteins. Some days a bit more, some a bit less. If that feels excessive, and you find yourself dreading eating these foods, scale it back a bit.
- Plenty of fiber, vitamins, and minerals and micronutrients, from vegetables: 35 grams of fiber recommended for men each day.
- 25 grams of fiber for women each day.
- Healthy fats: Like olive oil, avocados, and maybe a bit from dairy like Greek or (my favorite) Bulgarian yogurt or nuts. Small amounts of butter are fine. Nuts and nut butters are calorically dense, and most people tend to overeat them, but assuming they contain no added oils or sugars (including cane sugar) they are also fine. (Read the packaging; manufacturers seem to sneak those into everything nowadays!)
- Fruit: Two to four pieces or servings per day depending on activity levels and caloric needs. Berries are a great choice, but expensive. Look for citrus, apples, peaches, and on and on. If you want to look up the sugar and fiber content of different fruit, do; opt for the lower sugar, higher fiber varieties if you can.

- Clean starches: Think rice, oatmeal, potatoes, yams, sourdough or rye bread, beans, lentils. The goal is to eat enough of these to fuel your brain and bodily activities, including regular moderate to intense exercise four to six days per week.

That's what to seek out.

And not to sound like a broken record, but most people are best off limiting starch-fat combination foods and starch-fat-sugar combination foods. That means very little, if any, highly processed food like candy or cookies, boxed or frozen meals, or sugary drinks. Just cut them out. You'll feel so much better in every way—not just in appearance, but your cravings for sugar will be greatly reduced.

If you want a decadent homemade or real bakery-made dessert, homemade pasta, or a couple of slices of pizza or ice cream now and again—as in up to once a week—no big deal (especially if you're exercising as described in chapter 2).

Alas, there is no official name for this diet. While it's pretty straightforward, it's certainly not the Standard American Diet. Nor is it paleo or Mediterranean or animal-based. That's why I called it the Clean Omnivore Diet. It's clean, it's healthy, and it's satiating. It's also flexible, meaning it works both for your preferences and your goals. For instance, at the end of this section I offer a suggestion that

explains how one slight modification of this diet can allow for meaningful fat loss that you can maintain over time.

So aim for 30 to 50 grams of high-quality protein at each meal. That's about one large chicken breast, a 6- to 8-ounce piece of salmon or other fish, or a 5- to 7-ounce steak. This will fill about a third of a standard dinner plate.

Fill another third of that plate with vegetables.

If you're having starch with that meal, the starch can fill the remaining third of the plate. This is approximate. Don't obsess. If you completed an intense resistance training session or a hard cardio training session lasting longer than forty-five minutes that day, feel free to increase the starch portion a bit. Otherwise, keep your starch intake moderate.[44]

HOW MUCH TO EAT: CALORIES IN, CALORIES OUT

Whatever diet you decide is best for you, you'll need to pay attention to how many calories you consume. Calorie is the unit we use to measure how much energy is in what we eat and drink. And the number of calories you consume versus the number of calories you burn determines whether you will gain weight, lose weight, or stay at your current weight. "Calories in, calories out" is the phrase to remember. Or, if

[44] Vicente Javier Clemente-Suárez et al., "The Burden of Carbohydrates in Health and Disease," Nutrients 14, no. 18 (2022): 3809, https://doi.org/10.3390/nu14183809.

you prefer acronyms, CICO. This is not a matter for debate; it's a matter of thermodynamics.

If you want to lose weight, then you must consume (and absorb) fewer calories than you burn. I say "absorb" because not every calorie you consume is absorbed. Complete proteins are 4 calories per gram as they head into your mouth, but the metabolic requirement of processing those proteins ends up making them closer to 3 calories per gram when all is said and done. And while it's not a dinner-table-appropriate topic, some of the calories in food pass through your digestive tract without being absorbed. For example, humans have engineered fats that improve the taste of a food, but that get eliminated from your digestive tract with no absorption of calories. Other calories fail to be absorbed because they have a rapid rate of passage through your digestive tract, or bind to fiber. But for the most part, calories eaten equals calories absorbed.

Which takes us to the calories-out part of CICO.

As covered in chapter 2, your basal metabolism is the number of calories you burn if you were to just lie in bed all day and night and do literally nothing. Contrary to what many people once believed, your basal, or resting, metabolism doesn't change a whole lot with age, at least not

after age twenty-five or so.[45] Unless your muscle mass changes. If you lose muscle, your resting metabolic rate decreases. If you gain muscle, your resting metabolic rate will increase. This suggests that one reason people put on weight as they age is they lose muscle mass. Another is that as people get older in age, they tend to walk less and just generally move around less (even while seated!) and thus burn fewer calories each day. This kind of non-exercise movement, or non-exercise activity thermogenesis (NEAT) in nerd-speak, may not seem important, but it is. It can burn greater than 1,000 calories a day in some people… so knee bouncers and fidgeters rejoice.[46]

While NEAT is (amazingly) a key aspect to your daily caloric expenditure, you of course can't get away with just fidgeting and walking. As explained in detail in chapter 2, exercise—specifically resistance training and cardiovascular training—is critical to maintaining healthy body composition over time. Also, if you are choosing to be in a calorie deficit to lose weight, and you resistance train, more of the weight you lose will come from body fat and less will come from

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[45] Herman Pontzer et al., "Daily Energy Expenditure Through the Human Life Course," Science 373, no. 6556 (August 2021): 808–12, https://doi.org/10.1126/science.abe5017.

[46] Christian von Loeffelholz and Andreas L. Birkenfeld, "Non-Exercise Activity Thermogenesis in Human Energy Homeostasis," in Endotext (Internet), ed. K. R. Feingold et al. (MDText.com, 2022), https://www.ncbi.nlm.nih.gov/sites/books/NBK279077/.

muscle. If you don't resistance train, you'll lose roughly equal amounts of body fat and muscle.

The reverse is also true. If you're in a calorie excess, you'll gain weight that is a mix of muscle and body fat, but probably more body fat. That is, unless you resistance train, in which case you'll gain more weight that is muscle, especially if many of the additional calories are from high CP:C ratio proteins.[47] And if you resistance train while eating at maintenance levels, you'll likely add muscle and reduce your overall body fat percentage.[48]

So viewed purely through the lens of thermodynamics, does it matter what you eat? If you are thinking about total weight on the scale, then technically no, it doesn't matter. A calorie is a calorie. But as you can see from everything we've discussed so far in the chapter, there's more to it than that.

Physiologically speaking, we eat for four main reasons:

- To provide our cells with the raw materials like amino acids and fatty acids that they need to rebuild themselves
- To provide our cells with glucose to fuel their activities, that is, to give them energy

[47] Gary John Slater et al., "Is an Energy Surplus Required to Maximize Skeletal Muscle Hypertrophy Associated with Resistance Training?" Frontiers in Nutrition 6 (August 2019): 131, https://doi.org/10.3389/fnut.2019.00131.

[48] Alex S. Ribeiro et al., "Effects of Different Dietary Energy Intake Following Resistance Training on Muscle Mass and Body Fat in Bodybuilders: A Pilot Study," Journal of Human Kinetics 70, no. 1 (November 2019): 125–34, https://doi.org/10.2478/hukin-2019-0038.

- To get micronutrients (such as vitamins and minerals) for our cells
- To provide sustenance for the beneficial bacteria that live in our gut

(We also eat for psychological reasons, but those reasons are countless in number: because it's social, or it reminds us of home, or it reduces our anxiety, or it just tastes good, and on and on.)

That's why I believe, given most people are (and want to remain) omnivores, that the healthiest possible way to eat is to get enough high CP:C ratio proteins, plenty of fibrous vegetables and some fruit, healthy monounsaturated and polyunsaturated fats, and a bit of saturated fat, too, and enough clean starch to fuel our brains and bodies. And to do all that without eating more calories than we need. It hits each of these four reasons, while helping us stick to the requirements of CICO.

A Word About Weight Loss, or the Clean and Lean Omnivore Diet

One of the best ways to avoid eating too many calories is to avoid processed foods entirely, while also restricting starch-fat and starch-fat-sugar combinations, even if the ingredients are not processed.

There is even a case for dropping starches entirely—at least for a while. I know it seems crazy, but I have a dozen or more close friends who really struggled with their weight for years who switched to a diet consisting of meat, fish, eggs, vegetables, fruit, olive oil, coffee, tea, and water (there's an argument to be made that diet soda is fine to include too) and only used salt, pepper, and cinnamon for added flavoring.[49] Without any calorie counting, they have lost anywhere from twenty-five to sixty pounds in four to six months, and kept it off. These friends also made sure to consume some low-sugar fermented foods each day, too, such as sauerkraut or kimchi, which promote microbiome health. All of them exercised but no more than they were doing prior to eating this way. Each reported "cheating" two to three times a year, usually for a holiday meal or birthday. They simply got right back to the plan the next day. And after a year or so of eating this way, the ones exercising regularly and intensely were able to start adding moderate amounts of clean starches, like potatoes, beans, and rice, back into their regimen with no detriment.

Their results were remarkable, but not atypical. Again, since the laws of thermodynamics are immutable, we have to conclude this diet naturally puts active people into a caloric

[49] John C. Peters et al., "The Effects of Water and Non-Nutritive Sweetened Beverages on Weight Loss and Weight Maintenance: A Randomized Clinical Trial," Obesity 24, no. 2 (December 2015): 297–304, https://doi.org/10.1002/oby.21327.

deficit. What is wild is how fast it works, and how easy it is to stick to, and how it stands in bold contrast to their many other failed attempts to lose body fat in a sustainable way. You might say “No bread?! No ice cream?! No pasta?!” All I know is that the people who elected to eat this way report feeling very sated, which I attribute to the fact that they were getting everything their body needed physiologically and psychologically, without counting calories. And their improved body composition and other indicators of health, including sleep, just further reinforced their decision. I guess if we had to name this diet we would call it the Clean Omnivore No Starch Diet but that’s a mouthful so I’ve defaulted to calling it the Clean and Lean Omnivore Diet instead.

Calculating Your CICO

After you’ve decided on what to eat, you’ll want to dial in to how much. That’s where calculating your CICO comes in.

I actually think it’s a good idea for people to calculate their CICO for a few days or so, at least one to two times a year, and update their food intake accordingly, the same way that I think it’s a good idea to get your blood pressure checked one to two times a year, and update your stress control tools accordingly.

Luckily there is an easy, no-cost way to determine your caloric needs. It's called the Katch-McArdle formula. I discussed it in chapter 2, so please refer to page 174 to refresh your memory on how your daily caloric needs are calculated.

If you want additional support for body fat loss, muscle gain, et cetera, as I mentioned I highly recommend the Carbon App, created by Layne Norton, one of the world's foremost experts on nutrition, especially for health and body composition. (Note: Carbon is a sponsor of my podcast.) It allows you to determine what you need to eat and how much of each macronutrient and it makes it easy to enter your calorie and specific food intake. It also learns your nutrition patterns over time, and makes valuable suggestions. So it's not just a calorie and macronutrient counter; it's a nutrition coach. I have used it for years now and everyone I have recommended it to thanks me for the ease and results it's brought them around nutrition and body composition.

STRAIGHT TALK ABOUT HIGHLY PROCESSED FOODS

It is true that eating 3,000 calories of highly or "ultra" processed food will not cause any more or less weight gain than eating 3,000 calories of non-processed food, but the damage the former does is real. Let's acknowledge the one thing everyone in the nutrition field seems to agree on—the

vegans, the carnivores, the keto enthusiasts, the vegetarians, and the omnivores: Highly processed foods are genuinely bad for us.

When we talk about highly processed foods, we're referring to multi-ingredient, packaged products that are either ready-to-eat or only need to be heated up to eat. We're not referring to single ingredient starches that just need to be baked or cooked in boiling water, like rice or potatoes or beans. That's not "processing"; it's "preparation": And those are healthy, non-processed foods. Highly processed foods are problematic for many reasons related to our short- and long-term health. These products—even the ones labeled "Heart Healthy"—are often high in sugar, as well as various preservatives and artificial dyes that, while perhaps not deadly on their own, seem to pose a risk to health.[50]

And recent studies suggest that eating a lot of processed foods can increase your risk of psychiatric illnesses, likely due to changes in brain metabolism.[51] In fact, that is one reason why, as mentioned above, ketogenic diets (which are very low in carbohydrates) are increasingly being prescribed as a tool to improve mental health. Whether that is due to the

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[50] G. Pagliai et al., "Consumption of Ultra-Processed Foods and Health Status: A Systematic Review and Meta-Analysis," British Journal of Nutrition 125, no. 3 (2021): 308–18, https://doi.org/10.1017/S0007114520002688.

[51] Melissa M. Lane et al., "Ultra-Processed Food Consumption and Mental Health: A Systematic Review and Meta-Analysis of Observational Studies," Nutrients 14, no. 13 (2022): 2568, https://doi.org/10.3390/nu14132568; Lane et

absence of carbohydrates, the reduced intake of processed foods, or a direct effect of improved insulin sensitivity remains unclear, but the results are nonetheless compelling. We also have reliable evidence that diets low in fiber and high in processed foods, or even simply just high in refined carbohydrates, such as white rice and white wheat flour, increase the risk of cancer[52] and cardiovascular disease.[53] As do deli meats, which are effectively highly processed meats.[54]

It's hard to tack the health-degrading effects of processed foods to one aspect of their makeup. A study published in

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al., "Ultra-Processed Food Exposure"; Huiping Li et al., "Association of Ultraprocessed Food Consumption with Risk of Dementia: A Prospective Cohort Study," Neurology 99, no. 10 (2022): e1056–66, https://doi.org/10.1212/WNL.0000000000200871; Shenghao Yuan et al., "Associations of Ultra-Processed Food Intake and Its Circulating Metabolomic Signature with Mental Disorders in Middle-Aged and Older Adults," Nutrients 17, no. 9 (2025): 1582, https://doi.org/10.3390/nu17091582.

[52] Ryan J. Koene et al., "Shared Risk Factors in Cardiovascular Disease and Cancer," Circulation 133, no. 11 (2016): 1104–14, https://doi.org/10.1161/CIRCULATIONAHA.115.020406.

[53] Sonia S. Anand et al., "Food Consumption and Its Impact on Cardiovascular Disease: Importance of Solutions Focused on the Globalized Food System: A Report from the Workshop Convened by the World Heart Federation," Journal of the American College of Cardiology 66, no. 14 (2015): 1590–1614, https://doi.org/10.1016/j.jacc.2015.07.050.

[54] Véronique Bouvard et al., "Carcinogenicity of Consumption of Red and Processed Meat," Lancet Oncology 16, no. 16 (2015): 1599–600, https://doi.org/10.1016/S1470-2045(15)00444-1; Renata Micha et al., "Red and Processed Meat Consumption and Risk of Incident Coronary Heart Disease, Stroke, and Diabetes Mellitus: A Systematic Review and Meta-Analysis," Circulation 121, no. 21 (2010): 2271–83, https://doi.org/10.1161/CIRCULATIONAHA.109.924977.

2017 reported that nearly half of all deaths due to cardiac or metabolic issues are rooted in poor diet.[55] The Center for Science in the Public Interest, a nonprofit organization, estimates that 678,000 US deaths each year are due to diseases stemming from an inadequate or otherwise poor diet.[56] Half! Personally, I view highly processed foods as destructive to health mostly because of the other actually healthy foods you could be eating in their place. You still have to obey CICO according to your goals, and make sure you get the right macro- and micronutrients, but avoiding processed foods will likely add years to your life.

WHEN TO EAT

Then there is the question of when we eat. Here again, people debate. In the 1990s it became popular to eat four to six small meals per day, following a 1989 study published in the New England Journal of Medicine on the subject of "nibbling versus gorging."[57] The idea was that grazing throughout the day would prevent overeating in any one sitting and stimulate metabolism. It fell out of favor because

[55] Renata Micha et al., "Association Between Dietary Factors and Mortality from Heart Disease, Stroke, and Type 2 Diabetes in the United States," JAMA 317, no. 9 (2017): 912–24, https://doi.org/10.1001/jama.2017.0947.

[56] "Why Good Nutrition Is Important," Center for Science in the Public Interest, n.d., https://www.cspi.org/eating-healthy/why-good-nutrition-important.

[57] David J. A. Jenkins et al., "Nibbling Versus Gorging: Metabolic Advantages of Increased Meal Frequency," New England Journal of Medicine 321, no. 14 (1989): 929–34, https://doi.org/10.1056/NEJM198910053211403.

it's impractical. It also turned out to have a trivial impact on metabolism.[58]

Most people default to eating three to four times a day, with one to two of those instances as snacks and the others as meals. Job, school, and social constraints usually dictate when, exactly, people eat.

The practice of time-restricted eating, also called intermittent fasting, as discussed earlier in this chapter, became popular a few years back, and many people follow this practice either deliberately or by default because they are not hungry in the morning. They end up restricting their eating to a window of four to ten hours each day, and most typically seven to eight hours.[59] Some people limit themselves to one meal a day, which means their window is only one to two hours a day. There is now a fair bit of science—including clinical trials—on time-restricted eating. Intermittent fasting was made popular first as a way to lose weight and improve insulin sensitivity, meaning the extent to which our cells use glucose for energy (the more efficiently this happens, the less insulin we need to produce in order to regulate blood

[58] Jameason D. Cameron et al., "Increased Meal Frequency Does Not Promote Greater Weight Loss in Subjects Who Were Prescribed an 8-Week Equi-Energetic Energy-Restricted Diet," British Journal of Nutrition 103, no. 8 (2010): 1098–1101, https://doi.org/10.1017/S0007114509992984.

[59] Ruth E. Patterson et al., "Intermittent Fasting and Human Metabolic Health," Journal of the Academy of Nutrition and Dietetics 115, no. 8 (2015): 1203–12, https://doi.org/10.1016/j.jand.2015.02.018.

sugar levels). Soon it was studied and touted as a tool to live longer, to avoid cancer, and on and on.[60]

When it comes to losing weight, the data show that intermittent fasting does indeed help when compared to "unrestricted eating." However, it likely helps because people end up eating fewer calories when they limit their eating to a specific time window. So ultimately, it all comes back to CICO. When you put intermittent fasting head-to-head with a diet where people can eat whenever they want, but the two diets are matched for total calories consumed, the results end up being the same. A yearlong randomized study found that people with obesity who followed a calorie-restricted diet and ate only within an eight-hour window did not lose more weight or body fat compared to those who followed calorie restriction without the accompanying time-restriction plan.[61] That being said, many people find it easier to control the hours in which they eat than to control the portion of what they eat, and thus find intermittent fasting a very effective tool for eating fewer calories than they would otherwise.

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[60] Sally Adee, "Does Intermittent Fasting Live Up to the Hype?," New York Times, January 26, 2026, https://www.nytimes.com/2026/01/26/well/eat/intermittent-fasting.html.

[61] Deying Liu et al., "Calorie Restriction with or Without Time-Restricted Eating in Weight Loss," New England Journal of Medicine 386, no. 16 (2022): 1495–504, https://doi.org/10.1056/NEJMoa2114833.

Many people opt for intermittent fasting because they incorrectly think that means you don't have to think about the types or amount of food you eat, but if you care about your health you still do. Fasting doesn't change our need for micronutrients.

Debate continues about what it can do. Intermittent fasting has generated interest for its potential to extend life.[62] However, the only strong link between fasting and extended lifespan appears to be restricting calories, not meal timing.[63] That said, Satchin Panda's lab at the Salk Institute has shown that mice who eat across a twenty-four-hour window of time compared to an eight-to-twelve-hour window, even when calories are held constant, suffer negative health outcomes such as obesity, elevated glucose levels, insulin resistance, and fatty liver.[64] Studies from Satchin's lab have also shown similar adverse effects for humans.[65] Further studies show

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[62] Fran Smith, "Can Fasting Help You Live Longer? Here's What the Science Says," National Geographic, January 17, 2023, https://www.nationalgeographic.com/magazine/article/can-fasting-help-you-live-longer-what-the-science-says.

[63] Arthur V. Everitt and David G. Le Couteur, "Life Extension by Calorie Restriction in Humans," Annals of the New York Academy of Sciences 1114, no. 1 (2007): 428–33, https://doi.org/10.1196/annals.1396.005.

[64] Megumi Hatori et al., "Time-Restricted Feeding Without Reducing Caloric Intake Prevents Metabolic Diseases in Mice Fed a High-Fat Diet," Cell Metabolism 15, no. 6 (June 2012): 848–60, https://doi.org/10.1016/j.cmet.2012.04.019.

[65] Mark P. Mattson et al., "Meal Frequency and Timing in Health and Disease," Proceedings of the National Academy of Sciences 111, no. 47 (November 2014): 16647–53, https://doi.org/10.1073/pnas.1413965111.

that not eating within two to three hours of bedtime does seem to improve sleep quality,[66] growth hormone release[67] in sleep, and next-morning glucose regulation,[68] all of which may indirectly impact lifespan. While restricting eating within two to three hours of bedtime does seem to be a best practice, there are some exceptions based on the type of meal. For instance, if you're hungry and need a snack, consuming 30 grams of casein, the protein found in dairy, can help reduce blood pressure and perhaps even aid in exercise recovery, especially for those doing resistance training.[69]

Some research suggests that time-restricted eating extends life because it increases autophagy, a process that removes damaged or used cells and cellular material from our

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[66] Li-Ming Yan et al., "Chronobiological Perspectives: Association Between Meal Timing and Sleep Quality," PLOS One 19, no. 8 (2024): e0308172, https://doi.org/10.1371/journal.pone.0308172.

[67] Gabrielle Brandenberger and Laurence Weibel, "The 24-h Growth Hormone Rhythm in Men: Sleep and Circadian Influences Questioned," Journal of Sleep Research 13, no. 3 (2004): 251–55, https://doi.org/10.1111/j.1365-2869.2004.00415.x.

[68] Marie-Pierre St-Onge et al., "The Interrelationship Between Sleep, Diet, and Glucose Metabolism," Sleep Medicine Reviews 69 (June 2023): 101788, https://doi.org/10.1016/j.smrv.2023.101788.

[69] Arturo Figueroa et al., "Effects of Milk Proteins and Combined Exercise Training on Aortic Hemodynamics and Arterial Stiffness in Young Obese Women with High Blood Pressure," American Journal of Hypertension 27, no. 3 (March 2014): 338–44, https://doi.org/10.1093/ajh/hpt224.

bodies.[70] We also know that extended fasting, such as consuming only water and electrolytes for one to three days, is linked to increased autophagy.[71] But again here, it's not the timing that increases autophagy; it's the overall daily calorie deficit.[72] And whether autophagy truly extends human lifespan is still unclear.[73] Most evidence we have linking caloric restriction or intermittent fasting to lifespan is still largely from animal studies.[74] Prolonged fasting in humans will no doubt improve certain health metrics, like insulin sensitivity, but it risks reducing lean mass. Exercise—both resistance and cardio training—stimulate autophagy but improve lean mass with the added benefit of improving fitness.[75]

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[70] Matt Ulgherait et al., "Circadian Autophagy Drives iTRF-Mediated Longevity," Nature 598 (2021): 353–58, https://doi.org/10.1038/s41586-021-03934-0.

[71] Sebastian Brandhorst et al., "A Periodic Diet That Mimics Fasting Promotes Multi-System Regeneration, Enhanced Cognitive Performance, and Healthspan," Cell Metabolism 22, no. 1 (2015): 86–99, https://doi.org/10.1016/j.cmet.2015.05.012.

[72] Ki Wung Chung and Hae Young Chung, "The Effects of Calorie Restriction on Autophagy: Role on Aging Intervention," Nutrients 11, no. 12 (2019): 2923, https://doi.org/10.3390/nu11122923.

[73] Frank Madeo et al., "Essential Role for Autophagy in Life Span Extension," Journal of Clinical Investigation 125, no. 1 (2015): 85–93, https://doi.org/10.1172/JCI73946.

[74] Richard Weindruch and Rajindar S. Sohal, "Caloric Intake and Aging," New England Journal of Medicine 337, no. 14 (1997): 986–94, https://doi.org/10.1056/NEJM199710023371407.

[75] Jens Frey Halling and Henriette Pilegaard, "Autophagy-Dependent Beneficial Effects of Exercise," Cold Spring Harbor Perspectives in Medicine 7, no. 8 (March 2017): a029777, https://doi.org/10.1101/cshperspect.a029777.

I reflexively do something similar to intermittent fasting. My first meal is typically around 11:00 a.m. or noon. I wake up at about 6:00 a.m. most days and I find it easy to wait to eat my first meal because I tend not to get hungry until 10:30 or so. Around 3:30 p.m. I usually get hungry again and I'll have a snack. I prefer to eat my last meal of the day around 7:00 p.m., but I stay somewhat flexible on that depending on my schedule.

Since I'm typically done eating around 8:00 p.m., my eating window is usually eight to nine hours, maybe a bit longer. I've found that this slightly restricted window allows me to feel at my best while also helping me to manage my daily caloric intake.

From a health standpoint, it's best to eat at fairly consistent times each day,[76] but you don't have to be neurotic about it. If someone sets a dinner time for an hour earlier or later than usual, or a bit outside your eating window, it's not a huge deal.

WHAT'S AHEAD

The following protocols are not yet another trend to follow. When I said I hoped you'd find the best diet for you, what I meant is one that you can stick to.

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[76] Esther Ugo Alum, "Circadian Nutrition and Obesity: Timing as a Nutritional Strategy," Journal of Health, Population and Nutrition 44, no. 1 (2025): 367, https://doi.org/10.1186/s41043-025-01102-y.

These protocols are not complicated, they are not expensive, and they do not involve searching out obscure ingredients or creating an elaborate schedule of meal prep. They follow directly from the ideas discussed in this introduction to the chapter. Each explores a key point here, then offers one simple tool. My hope is that they will provide you with the inspiration and rationale to make easy yet meaningful shifts in your nutrition, and therefore immediate and long-term health. Adopting any of these into your daily routine can improve your digestion, your energy levels, and your ability to focus and concentrate. For those of you seeking dramatic positive transformations in body composition, I offer a sustainable way to do that too. Of course, exercise—per the recommendations in chapter 2—is critical for getting sustainable and healthy body composition changes, and exercise has its own benefits altogether. Some of the suggestions that follow will be familiar. Others may sound a bit heretical. But I assure you they work.

1

NUTRITION PROTOCOL

90 Percent or More of the Food You Eat Should Be Unprocessed or Minimally Processed

WHAT TO DO

- Eat and cook with single-ingredient, non-processed foods. These include meat, fish, eggs, poultry, fruit, vegetables, rice, lentils, beans, oatmeal, pasta, olive oil, avocado oil, and even butter (albeit in small portions). They can be consumed alone or combined in recipes, and of course you can use spices, seasonings, and salt as well.
- Avoid ready-to-eat packed foods or packaged multi-ingredient foods. Most pre-packaged sauces and condiments should be avoided; these typically contain a fair amount of added sugar. Read nutrition labels to check.
- Consume nut butters in moderation. These can be healthy, but some contain processed oils and even added sugar. Even when nuts and salt are the only ingredients, all have a very high calorie-to-protein ratio. If you're trying to lose body fat, then it's best to abstain from nut butters entirely.
- Consume full-fat yogurt and cottage cheese and other forms of dairy, but in plain flavors only. Here too added sugars should be avoided. Add fruit, a drizzle of honey,

or a pinch of cinnamon for taste if you don't like these foods plain.

HOW IT WORKS

Food is amazing. Despite the debates, I think we can all agree on that. We eat for the building blocks food supplies our brain and body, as described in the introduction to this chapter, and of course for the pleasure of it. For many people, food is one of their biggest joys. I count myself among people who love to eat.

But the type of food we eat is probably second only to sleep and exercise for its importance to our long-term health. Centering our meal and snack choices on whole (that is minimally or non-processed) foods has so many benefits—disease prevention, longevity, cognition, mood, energy, reduced inflammation, healthy weight management, and more. The increasing concerns about processed foods stem from the plethora of data linking them to obesity, high blood pressure, heart disease, gastrointestinal disorders, cancer, and other serious ailments.[77] Chemical additives used to color, preserve, texturize, and flavor these products have been linked to potential health hazards.[78] Processed foods also tend to contain high amounts of sugar, even in

[77] Monteiro et al., "Ultra-Processed Foods."

[78] National Research Council (US) Committee on Diet, Nutrition, and Cancer, Diet, Nutrition, and Cancer: Directions for Research (National Academies Press [US], 1983), chap. 8, https://www.ncbi.nlm.nih.gov/books/NBK216714/.

products that would seem to not require it. And excessive sugar consumption is tied to diabetes, heart disease, cancer, and an increased risk of early death.[79] All of which is to say, we need to make it a point to eliminate, or at least vastly reduce, processed foods.

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[79] Djésia Arnone et al., "Sugars and Gastrointestinal Health," Clinical Gastroenterology and Hepatology 20, no. 9 (2022): 1912–24.e7, https://doi.org/10.1016/j.cgh.2021.12.011; Fumiaki Imamura et al., "Consumption of Sugar Sweetened Beverages, Artificially Sweetened Beverages, and Fruit Juice and Incidence of Type 2 Diabetes: Systematic Review, Meta-Analysis, and Estimation of Population Attributable Fraction. BMJ 351 (2015): h3576, https://doi.org/10.1136/bmj.h3576; Vasanti S. Malik et al., "Sugar-Sweetened Beverages and Risk of Metabolic Syndrome and Type 2 Diabetes: A Meta-Analysis," Diabetes Care 33, no. 11 (2010): 2477–83, https://doi.org/10.2337/dc10-1079; Bo Yang et al., "Added Sugar, Sugar-Sweetened Beverages, and Artificially Sweetened Beverages and Risk of Cardiovascular Disease: Findings From the Women's Health Initiative and a Network Meta-Analysis of Prospective Studies," Nutrients 14, no. 19 (October 2022): 4226, https://doi.org/10.3390/nu14204226; Quanhe Yang et al., "Added Sugar Intake and Cardiovascular Diseases Mortality Among US Adults," JAMA Internal Medicine 174, no. 4 (April 2014): 516–24, https://doi.org/10.1001/jamainternmed.2013.13563; Florida Llaha et al., "Consumption of Sweet Beverages and Cancer Risk: A Systematic Review and Meta-Analysis of Observational Studies," Nutrients 13, no. 2 (2021): 516, https://doi.org/10.3390/nu13020516; Yuting Li et al., "Consumption of Sugar-Sweetened Beverages and Fruit Juice and Human Cancer: A Systematic Review and Dose-Response Meta-Analysis of Observational Studies," Journal of Cancer 12, no. 10 (2021): 3077–88, https://doi.org/10.7150/jca.51322; Yan-Bo Zhang et al., "Association of Consumption of Sugar-Sweetened Beverages or Artificially Sweetened Beverages With Mortality: A Systematic Review and Dose-Response Meta-Analysis of Prospective Cohort Studies," Advances in Nutrition 12, no. 2 (2021): 374–83, https://doi:10.1093/advances/nmaa110; Lukas Schwingshackl et al., "Food Groups and Risk of All-Cause Mortality: A Systematic Review and Dose-Response Meta-Analysis of Prospective Studies," American Journal of Clinical Nutrition 105, no. 6 (2017): 1462–73, https://doi.org/10.3945/ajcn.117.153148.

The good news is that eating healthily is not complicated or expensive. We can all make incredible tasting meals and snacks using whole foods, if we learn just a bit about how to prepare them. Think: a perfectly cooked steak and carrots roasted with olive oil, salt, and some spices. In some cases, these foods need no preparation at all. Like a juicy orange or fresh berries. If these sound bland or overly simple, chances are your taste circuits in your brain and body need some recalibration.

If you have grown accustomed to eating processed foods, this recalibration can seem difficult. Processed foods alter our brain and gut to make us perceive other, more nutritious foods as less appetizing. It's because of how sugar, certain starch-sugar, and starch-sugar-fat combinations are metabolized in our gut, and how they change our brain's reward pathways. You'll be happy to know, however, that the cells in your gut and brain are perfectly happy to rewire to make healthy, unprocessed foods literally taste better. It doesn't take long, and it will make eating healthy most of the time reflexive.

This protocol looks at the gut-brain axis, the connection between the cells in our digestive tract and specific areas of our brain. That connection is the key to understanding how processed foods affect us.

Our gut talks to our brain via two main parallel pathways. The first involves the cells in our gut communicating with our brain using a neural pathway called the vagus nerve, a bidirectional superhighway linking the organs of our body and our brain. You may remember the vagus from chapter 3.

Our body and our brain are connected by a second pathway, through secreted hormones and small proteins called peptides (in general, the vagus pathway is the faster of the two). The hunger sensation arises, in part, because our stomach produces a hormone called ghrelin[80] in amounts proportional to how much time has elapsed since our last meal. Ghrelin travels to our hypothalamus, a marble-sized brain region located over the roof of our mouth, where it activates our drive to seek food.[81] Other hormones and peptides made by both our gut and our brain make us want to eat less; for example, the peptide GLP-1 gives us a feeling of fullness, which is why GLP-1 analogs are used as drugs to treat obesity. (The gut microbiome, the trillions of bacteria living in our digestive system, also influences hunger—more on that in Nutrition Protocol 6.)[82]

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[80] Jon Davis, "Hunger, Ghrelin and the Gut," Brain Research 1693, Pt. B (2018): 154–58, https://doi.org/10.1016/j.brainres.2018.01.024.

[81] Emeran A. Mayer et al., "The Gut-Brain Axis," Annual Review of Medicine 73 (January 2022): 439–53, https://doi.org/10.1146/annurev-med-042320-014032.

[82] C. S. Byrne et al., "The Role of Short Chain Fatty Acids in Appetite Regulation and Energy Homeostasis," International Journal of Obesity 39, no. 9 (2015): 1331–38, https://doi.org/10.1038/ijo.2015.84.

At the same time, specific nutrients in the foods we eat—most notably sugars, but also fatty acids and amino acids—activate cells in our gut called neuropod cells that communicate with the brain via the vagus nerve, leading to the release of dopamine.[83] And while dopamine does many different things in the brain and body, when it is stimulated in this context, it can drive the urge to consume more of that food. It's an elegant mechanism if you think about it, because the taste of food in our mouth is not a great indication of the actual nutrients in a given food. Taste is more of a guess, a correlate. The cells in our gut, on the other hand, are poised to respond to the broken-down components of food, letting the rest of our body and brain know what we ate and what to eat more (or less) of. Amazingly, all this happens below our conscious awareness.

Given how sugar drives the urge to consume more, it's no mystery why it's in so many foods. Processed food manufacturers add sugar to foods so we buy and consume more of these foods. That's why sugar can be found in everything from packaged mac-and-cheese to frozen pizza,

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[83] Serika Yamada et al., "Immediate Glucose Signaling Transmitted via the Vagus Nerve in Gut–Brain Neural Communication," iScience 28, no. 5 (2025), https://doi.org/10.1016/j.isci.2025.112439; Melanie Maya Kaelberer et al., "Neuropod Cells: The Emerging Biology of Gut-Brain Sensory Transduction," Annual Review of Neuroscience 43 (July 2020): 337–53, https://doi.org/10.1146/annurev-neuro-091619-022657; Andrew Huberman, "Dr. Diego Bohórquez: The Science of Your Gut Sense & the Gut-Brain Axis," Huberman Lab (podcast audio), May 27, 2024, https://hubermanlab.com/episode/dr-diego-bohorquez-the-science-of-your-gut-sense-the-gut-brain-axis.

and most everything else on the supermarket shelves. Even tortilla and potato chips can have added sugar. So does most store-bought pasta sauce. And salad dressings. And condiments. Which is why so many people crave sugary food all the time.

You get the idea. It's a clever, albeit diabolical, tactic.

As mentioned in the introduction to this chapter, sugar that gets added to food is considered a simple carbohydrate. Typically it is added as sucrose, but sometimes as high-fructose corn syrup, or something similar. The exact form doesn't really matter however. Our brains are hardwired to detect simple carbohydrates. Studies in both mice and humans show that even when we are unable to taste sweetness, we still show a strong preference for sugary foods and beverages, indicating that the brain can detect sugar by a pathway other than conscious taste.[84]

Here's a simple and fun experiment you can try; it's something that Anna Lembke, MD, a world expert in dopamine and its role in craving and addiction, taught me. Eat one small piece of chocolate or one spoonful of your favorite ice cream. Both have sugar and fat, a combination known to trigger a lot of dopamine compared to eating either one alone. Pay attention to how it tastes, but then also notice

[84] Hwei-Ee Tan et al., "The Gut–Brain Axis Mediates Sugar Preference," Nature 580, no. 7804 (2020): 511–16, https://doi.org/10.1038/s41586-020-2199-7.

your thoughts as you swallow. You'll notice how quickly they move from "mmm, yum" to "I want more." The immediate sense of craving that overlaps with pleasure is the power of dopamine in action. Sugar, in particular, amplifies that response, even after the flavor experience has passed, by activating the gut-to-brain pathway.[85] It even works for sugary foods that don't taste that good! And it's why processed foods that we don't even think of as sweet often contain added sugar.

Again, I want to be clear that I do not subscribe to the view that all sugar is inherently bad. A homemade dessert, chocolate, or ice cream once in a while… all can definitely fit within a healthy diet. But I do think we should limit these foods and avoid, as much as possible, consuming non-dessert foods that have added sugar, as well as any highly processed foods, except on rare occasions or out of true necessity.

Maybe you're wondering about fruit, which as I mentioned contains a sugar called fructose. Fruit contains a lot of water, fiber, valuable nutrients, and polyphenols, compounds found

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[85] Nora D. Volkow et al., "Reward, Dopamine and the Control of Food Intake: Implications for Obesity," Trends in Cognitive Sciences 15, no. 1 (2011): 37–46, https://doi.org/10.1016/j.tics.2010.11.001; Ashley N. Gearhardt et al., "Social, Clinical, and Policy Implications of Ultra-Processed Food Addiction," BMJ 383 (October 9, 2023): e075354, https://doi.org/10.1136/bmj-2023-075354; Erratum, Gearhardt et al., "Social, Clinical, and Policy Implications," BMJ 383 (2023): 2679, https://doi.org/10.1136/bmj.p2679.

in plants that have various beneficial properties. All these components contribute to the prevention of heart disease, cancer, and other serious health issues.[86] And the high fiber content of most fruits slows down the absorption of sugar, which makes them less stimulating than desserts, sweets, or highly processed foods. But there are a few fruits that contain more fructose than others—like grapes. Note, too, that dried fruits, without their water content, now contain concentrated amounts of sugar. Fruit juices, like orange juice, are generally best avoided. They are high in sugar and calories, too, and the absence of fiber means that the sugar contained there is easily absorbed into the body, which means it spikes blood sugar quickly. Because that spike leads to a subsequent crash in blood sugar,[87] consuming it often can lead, in the short run, to fatigue, sugar cravings,[88] and, in the long term, an increased risk of diabetes[89] and heart disease.[90]

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[86] Hannah Cory et al., "The Role of Polyphenols in Human Health and Food Systems: A Mini-Review," Frontiers in Nutrition 5 (September 21, 2018): 87, https://doi.org/10.3389/fnut.2018.00087.

[87] Konstantinos Mantantzis et al., "Sugar Rush or Sugar Crash? A Meta-Analysis of Carbohydrate Effects on Mood," Neuroscience & Biobehavioral Reviews 101 (June 2019): 45–67, https://doi.org/10.1016/j.neubiorev.2019.03.016.

[88] Di Qin et al., "Sugar Addiction."

[89] Le Ma et al., "Beverage Consumption and Mortality Among Adults with Type 2 Diabetes: Prospective Cohort Study," BMJ 381 (April 19, 2023): e073406, https://doi.org/10.1136/bmj-2022-073406.

[90] Kristine K. Dennis et al., "Associations of Dietary Sugar Types with Coronary Heart Disease Risk: A Prospective Cohort Study," American Journal

Then there are artificial and low-calorie sweeteners. The data show that artificial and low-calorie sweeteners do not drive the same craving typical simple sugars do. Stevia for instance, is safe, and may even have some health benefits.[91] Aspartame appears to be safe too.[92] Saccharin and sucralose, though, may negatively impact the gut microbiome.[93] And there's interesting data on diet sodas. In a randomized controlled trial from 2024, people who consumed diet sodas lost more weight than people who drank equivalent amounts of water.[94] The underlying reasons include less hunger among soda drinkers and more cravings for sweetness among the water drinkers. At the same time, people who consume a lot of artificial sweeteners often experience increased overall appetite, perhaps due to glucose and insulin

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of Clinical Nutrition 118, no. 5 (2023): 1000–1009, https://doi.org/10.1016/j.ajcnut.2023.08.019.

[91] Victoria Peteliuk et al., "Natural Sweetener Stevia rebaudiana: Functionalities, Health Benefits and Potential Risks," EXCLI Journal 20 (September 22, 2021): 1412–30, https://doi.org/10.17179/excli2021-4211.

[92] "Aspartame Hazard and Risk Assessment Results Released," World Health Organization, July 14, 2023, https://www.who.int/news/item/14-07-2023-aspartame-hazard-and-risk-assessment-results-released.

[93] Alex Kidangathazhe et al., "Synthetic vs. Non-Synthetic Sweeteners: Their Differential Effects on Gut Microbiome Diversity and Function," Frontiers in Microbiology 16 (May 14, 2025): 1531131, https://doi.org/10.3389/fmicb.2025.1531131.

[94] Joanne A. Harrold et al., "Non-Nutritive Sweetened Beverages Versus Water After a 52-Week Weight Management Programme: A Randomised Controlled Trial," International Journal of Obesity 48 (2024): 83–93, https://doi.org/10.1038/s41366-023-01393-3.

disruption.[95] There isn't a broad rule that applies to all artificial and low-calorie sweeteners. For some people, they seem to help with weight loss, whereas for other people, they stimulate hunger in ways that may make it harder to keep from overeating. You have to figure out which group you are in.

If a few diet sodas a day help quell your sugar cravings at a time when you're trying to lose weight, then great. If you don't like diet sodas, there is no reason to consume them. I drink beverages containing stevia and occasionally enjoy an aspartame-sweetened diet soda. When developing the zero-sugar cold brew yerba maté for Mateina (as noted, a company I partially own), I chose to include a small amount of stevia. I drink that for much of the early part of the day. I avoid sucralose and saccharin.

COMMON QUESTIONS

Do you have any suggestions for beating sugar cravings

Yes. Some are conventional; one is not (more on that in the last protocol of this chapter). As with many cravings, sugar can be hard to quit. If you think you're overdoing it with sugar, then try committing to having no sugar for a certain period. I suggest one week. I promise that soon you will stop

[95] Sandhya P. Chakravartti et al., "Non-Caloric Sweetener Effects on Brain Appetite Regulation in Individuals Across Varying Body Weights," Nature Metabolism 7 (2025): 574–85, https://doi.org/10.1038/s42255-025-01227-8.

craving it so much, though it takes a few days to “get over the bump” (the heightened cravings) as they say.

Focus on consuming high CP:C ratio proteins, at least three times per day. And dietary fiber, mostly from vegetables and lower sugar fruits like apples and berries. If you crave chocolate, try and slowly develop a taste for raw cacao beans (five or six per day), which are rich in polyphenols, which again have various beneficial properties. Similarly, you can try 100 percent chocolate (not baking chocolate!). You read right. That’s no sugar, and no stevia either. It’s an acquired taste, but once people develop a taste for it, they generally like it. If you really struggle with sugar cravings, cut sugar out entirely but consider drinking a half tablespoon (about 3–5 g) of l-glutamine powder dissolved in water two to three times a day; this powder may reduce cravings by regulating blood sugar levels and thereby quelling the need for sugar.[96] This is a somewhat unconventional tool—one for which we need more studies—but anecdotally speaking (about my own experience), it works.

[96] Jerry R. Greenfield et al., “Oral Glutamine Increases Circulating Glucagon-Like Peptide 1, Glucagon, and Insulin Concentrations in Lean, Obese, and Type 2 Diabetic Subjects,” American Journal of Clinical Nutrition 89, no. 1 (2009): 106–13, https://doi.org/10.3945/ajcn.2008.26362.



NUTRITION PROTOCOL

Stay Hydrated

WHAT TO DO

Regardless of what and when you choose to eat:

- Consume 18 to 24 ounces of water upon waking. Try to do this first thing in the morning before drinking or eating anything else, including coffee or other forms of caffeine.
- Aim to consume about 80 ounces of water across the first ten hours of the day. That's about 2.4 liters. It is not necessary to consume 8 ounces each hour, but try to spread it out in a way that works for you. The earlier in the day you drink it the better. It's typically best to avoid a lot of water in the hours before bedtime, since you want to wake as infrequently as possible to use the bathroom.
- For optimal performance during exercise, consume water according to this equation: Your body weight in pounds divided by 30 equals the number of ounces to drink for every 15 to 20 minutes of your workout time. This calculation originates with the work of human performance expert Andy Galpin, director of the Human Performance Center at Parker University, and is referred to below as the "Galpin Equation." Example: Total body weight of 150 pounds ÷ 30 = 5 ounces every 15 to 20 minutes, which means 15 to 20 ounces consumed during

an hour-long workout. You can sip water throughout, drink at regular or non-regular intervals. Your pattern of consumption during exercise will of course be dictated by the type of exercise you're doing: running, cycling, swimming, et cetera.

HOW IT WORKS

Your body is made mostly of water. By one estimate, obtained by sending a weak electric current through the body, an adult human is about 62 percent water, with that percentage starting to decrease around sixty years of age.[97] Water is essential for every biological process that we need in order to live and function.[98] If there is one protocol that is most overlooked and that can immediately and significantly improve your energy levels, focus, and general feeling of well-being, it is this: Hydrate properly. It is rarely discussed. When we do hear about hydration, the message is typically a warning against dehydration, not the great benefits of hydration.

So let's talk about what hydration is, and why it's so powerful.

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[97] Hong Lu et al., "Body Water Percentage from Childhood to Old Age," Kidney Research and Clinical Practice 42, no. 3 (2023): 340–48, https://doi.org/10.23876/j.krcp.22.062.

[98] Joshua E. Brinkman et al., "Physiology, Body Fluids," StatPearls, last updated January 27, 2023, https://www.ncbi.nlm.nih.gov/books/NBK482447/.

Hydration is the process of providing the cells of our body with the water they need for all those biological processes. Some hydration is achieved by breathing, some is produced through metabolism, and some comes through the foods we eat. Most hydration comes from the liquids we consume. The water we take in through eating and drinking enters our gut and is absorbed into our bloodstream and finally our cells.

Proper hydration is crucial to both the body and the brain. Hydration helps prevent injuries, improves recovery, and is also important for maintaining body temperature during and following exercise.[99] It reduces the risk of kidney stones.[100] Drinking water stimulates the sympathetic branch of your autonomic nervous system, the circuitry that increases your alertness and focus. (See chapter 3 for more on this.) Research indicates that drinking water can trigger the release of norepinephrine, the hormone that helps keep us focused and alert.[101] In a study of fifteen children, proper hydration was tied to improved performance on tasks involving motor

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[99] Lawrence W. Judge et al., "Hydration to Maximize Performance and Recovery: Knowledge, Attitudes, and Behaviors Among Collegiate Track and Field Throwers," Journal of Human Kinetics 79 (July 2021): 111–22, https://doi.org/10.2478/hukin-2021-0065.

[100] Wisit Cheungpasitporn et al., "Treatment Effect, Adherence, and Safety of High Fluid Intake for the Prevention of Incident and Recurrent Kidney Stones: A Systematic Review and Meta-Analysis," Journal of Nephrology 29, no. 2 (2016): 211–19, https://doi.org/10.1007/s40620-015-0210-4.

[101] Jens Jordan et al., "The Pressor Response to Water Drinking in Humans: A Sympathetic Reflex?," Circulation 101, no. 5 (2000): 504–9, https://doi.org/10.1161/01.cir.101.5.504.

skills and memory.[102] Similar findings have been observed in adults.

Dehydration, on the other hand, quickly becomes problematic. It's also common. This is unfortunate because even slightly insufficient amounts of water in your body can diminish your concentration, alertness, short-term memory, visual perception, and other cognitive functions.[103] Mild dehydration can cause low-level confusion, fatigue, depression, and irritability.[104] One review found that a 2-percent decrease in body mass due to dehydration was enough to diminish mood, increase fatigue, and reduce alertness. That is remarkable because most people do not consume enough water to maintain hydration, and spend much of their day suffering the ill effects of mild dehydration… without even realizing it.

A mildly dehydrated state can reduce brain volume and alter the activity of neural pathways associated with motor output and cognition.[105] It also reduces blood flow, which in turn

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[102] Paula Booth et al., "Water Supplementation Improves Visual Attention and Fine Motor Skills in Schoolchildren," Education for Health Change in Learning & Practice 30, no. 3 (January 2012): 75–79.

[103] Barry M. Popkin et al., "Water, Hydration, and Health," Nutrition Reviews 68, no. 8 (2010): 439–58, https://doi.org/10.1111/j.1753-4887.2010.00304.x.

[104] DeAnn Liska et al., "Narrative Review of Hydration and Selected Health Outcomes in the General Population," Nutrients 11, no. 1 (2019): 70, https://doi.org/10.3390/nu11010070.

[105] Giuseppe Faraco et al., "Water Deprivation Induces Neurovascular and Cognitive Dysfunction Through Vasopressin-Induced Oxidative Stress,"

diminishes the supply of oxygen to the brain. What's more, it can cause headaches and lethargy, and of course a dry mouth and thirst.[106] It's important to note amidst all this that we can be dehydrated without registering thirst.

People engaging in intense exercise need to be especially vigilant about hydration, and not just in hot and/or dry environments either. In one study, eight trained cyclists rode for two hours on a stationary bike, consuming either 2.4 liters of an energy drink or 0.4 liters of water plus sports gel to equal the carbohydrates in the energy drink. The low-hydration riders became exhausted sooner and exerted less energy as they rode.[107] In a study of seven experienced resistance trainers (all male), insufficient hydration decreased performance during exercise.[108] These studies are small but are representative of what happens when we exercise.

Journal of Cerebral Blood Flow & Metabolism 34, no. 5 (2014): 852–60, https://doi.org/10.1038/jcbfm.2014.24.

[106] Naila A. Shaheen et al., "Public Knowledge of Dehydration and Fluid Intake Practices: Variation by Participants' Characteristics," BMC Public Health 18 (December 2018): 1346, https://doi.org/10.1186/s12889-018-6252-5.

[107] Tammie R. Ebert et al., "Influence of Hydration Status on Thermoregulation and Cycling Hill Climbing," Medicine & Science in Sports & Exercise 39, no. 2 (2007): 323–29, https://doi.org/10.1249/01.mss.0000247000.86847.de.

[108] Daniel A. Judelson et al., "Effect of Hydration State on Strength, Power, and Resistance Exercise Performance," Medicine & Science in Sports & Exercise 39, no. 10 (2007): 1817–24, https://doi.org/10.1249/mss.0b013e3180de5f22.

The quantities provided in this protocol do not need to be followed to the precise ounce or milliliter. They are general guidelines to help ensure sufficient hydration.

One reason to focus on getting about 80 ounces of fluid in the first ten hours of the day is that this time is when our kidneys function at their best. Our kidneys filter fluid and produce hormones to regulate water and salt balance, among other things, and their efficiency at doing so is strongly governed by our circadian clock[109] (see Sleep Protocol 1 for more on the circadian clock). Our kidneys are simply much better at filtering water and other fluids we drink in the morning and daytime than at night; their function tends to peak in midafternoon and is slowest around midnight.[110] Focusing most of your water intake to the time of day when fluid filtration by the kidneys is at its most efficient will also help prevent, or at least reduce, having to get up in the middle of the night to use the bathroom, which is among the more common causes of sleep disturbance.

The quantity of 80 ounces across the first ten hours of the day gives us a solid metric to aim for. Simply make sure to drink 8 to 16 ounces of water (one to two glasses) upon

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[109] Dmitri Firsov and Oliver Bonny, "Circadian Rhythms and the Kidney," Nature Reviews Nephrology 14 (2018): 626–35, https://doi.org/10.1038/s41581-018-0048-9.

[110] Jermaine G. Johnston and David M. Pollock, "Circadian Regulation of Renal Function," Free Radical Biology and Medicine 119 (May 1, 2018): 93–107, https://doi.org/10.1016/j.freeradbiomed.2018.01.018.

waking, and again approximately every waking hour during the day. That will get you in the ballpark of 80 ounces in the first ten hours. Individuals living in hot environments tend to need more water. So do people who move around a lot, even if that's in a temperature-controlled office or home.

But if you're exercising, that's where the Galpin Equation enters the picture. If you want to improve your exercise performance, hydrate according to this calculation. You'll be amazed by the difference it makes. Again, anyone living in hot environments will tend to need a bit more water here too; so increase the amount from the equation by a few extra ounces every fifteen to twenty minutes.

Beyond these three points, just remember: Stay hydrated. Hydration sounds boring but it's actually really powerful when it comes to improving health and performance and again, it's one of the more overlooked tools out there. It's important for your brain, for your skin and appearance, for your digestion, and for your mood. You will notice the difference in all these areas after starting to follow this protocol.

COMMON QUESTIONS

How do I know if I'm dehydrated

There are so many myths around this. Among the more popular ones is that clear urine indicates sufficient hydration.

Wrong. Many factors impact urine color besides hydration status. Your urine can be yellow and you can still be plenty hydrated. Another: If you pinch the skin on your hand, it should retract in less than two seconds. Wrong again. This response depends on age, skin thickness, and other factors. It's true that if you pinch the skin on the top of your hand and it does not return to its original state within three to five seconds, then there's a decent chance you're dehydrated. Same goes if your urine is much darker than usual. But that's not the best way to guide your hydration, nor is whether your ankles are swollen… all of these are myths.

The best way to be certain that you are hydrated is… to not wait for some less-than-reliable marker to tell you so. Simply drink enough water.

Do I follow both protocols at once when exercising? In other words, do I drink the recommended amount during general waking hours plus the amount recommended for hydration during exercise

Yes. Drink the amount recommended for ten hours of waking, and add more using the Galpin Equation. If you don't want to bother with the calculation, then just aim to ingest 80 ounces of water across the first ten hours of the day and add an extra 15 to 20 ounces of water for every hour spent exercising—and drink it while exercising.

Should I drink more if I'm someone who sweats a lot

People who tend to sweat a lot can increase the Galpin Equation by 50 to 100 percent during exercise. The same holds true if you are doing some kind of heat exposure that promotes sweating. For example, it would be appropriate to drink 8 to 16 ounces of fluid for every twenty to thirty minutes you spend in a hot sauna.

Electrolytes (particularly sodium, magnesium, and potassium) are also beneficial, especially if you are sweating. If you're working or you're exercising in a warm environment, or you're spending time in a hot sauna, add a pinch of sea salt to water (and add a little lemon juice to offset the salty taste). Or use a commercial source of electrolytes to help replace those lost through sweat. I'm a fan of LMNT, a sponsor of my podcast, which has more sodium than most electrolyte formulas. (It also has zero sugar, which is good since many hydration formulas have added sugar.) Many people worry about salt intake but if you eat clean—meaning limited or no processed foods—there is a good chance you need more, not less sodium.

Even if I slow down my water intake after the first ten hours of the day, I'm still concerned about frequent nighttime waking. Is there a way to reduce this

It's normal to wake up once during the night to urinate. To avoid waking more than once, try this:

- 1 Hydrate sufficiently during the day to ensure you are not excessively thirsty in the evening.
- 2 Reduce fluid intake in the evening (obviously).
- 3 If you do drink fluids (of any kind) later in the day or in the evening, sip them slowly. The rate at which fluid moves from the gut to the bladder is determined partly by the rate at which we drink. Gulping water quickly leads to sooner urination; sipping slowly leads to more time between urination.[111]

That said, if you are still waking up often, that could reflect a bladder or prostate issue. Talk to your doctor.

I drink coffee, tea, diet soda, et cetera. Does that count toward my daily hydration needs

Technically, yes, but drinking good ol' plain water is still important. Beverages containing caffeine act as diuretics; they stimulate urine production and therefore do not contribute to hydration to the same extent as non-caffeinated drinks. The amount of caffeine most people consume is usually not enough to cause significant dehydration, but anyone who drinks more than two cups of strong coffee per day—meaning more than 200 to 400 milligrams of

[111] Mohammad A. Shafiee et al., "Defining Conditions That Lead to the Retention of Water: The Importance of the Arterial Sodium Concentration," Kidney International 67, no. 2 (2005): 613–21, https://doi.org/10.1111/j.1523-1755.2005.67117.x.

caffeine—should make sure to drink at least those 80 ounces of water in addition, and perhaps as much as 100 ounces.

Keep in mind that many commercial coffee vendors brew their coffee strong, as in 300 milligrams or more of caffeine per serving. If you tend to feel shaky or lightheaded after drinking that coffee, you may be low on electrolytes, as caffeine causes excretion of sodium, a key electrolyte. I've known a lot of people who were able to eliminate caffeine-induced lightheadedness and shakes, and even the caffeine crash, by drinking a glass of water with a small pinch of salt dissolved in it, or by drinking water with a serving of electrolytes (sodium, magnesium, and potassium) alongside their coffee or other caffeine-containing drink. In fact, people who drink a lot of caffeinated beverages report that they feel much more energetic, and avoid the caffeine crash, when they are careful to also ingest some electrolytes before, alongside, or after their caffeine.



NUTRITION PROTOCOL

Eat High CP:C Ratio Protein at Each Full Meal

WHAT TO DO

- Aim to include 30 to 50 grams of high complete-protein-to-calorie ratio foods at each meal or snack. Your daily goal should be around 1 gram per

pound of body weight. This is the cornerstone of your diet.

- The number of calories from those foods will depend on what you select. Poultry, lean red meat, fish, and shellfish are the best high CP:C ratio sources of protein for those who eat animal-based products. Eggs are great as well, but they do have a relatively high calorie-to-protein ratio. Whey protein and plain full-fat Greek yogurt and cottage cheese too. Plant-based complete protein sources include beans and rice (together), lentils, soy, seitan, and some grains, such as quinoa. Just be aware that these have a less-than-favorable protein-to-calorie ratio.
- Nuts, nut butters, and most cheeses are not high-quality protein sources. And all of them are very calorie dense. Peanut, almond, and other nut butters are mostly fat, and most nut butters on the grocery shelf often also contain added sugar and/or seed oils. If you're going to eat nut butters, limit them to a few tablespoons a week and be sure they contain only nuts and salt. No sugar or added oils. It would be better to avoid nut butters and eat whole seeds and actual whole nuts. But these are mostly fats; they are not high CP:C ratio proteins.
- If you are someone who tends to overeat, whichever foods you choose, eat your high CP:C ratio proteins first. Then any starches, then your fibrous vegetables. You'll

enjoy more satiety, better appetite regulation, and better digestion because apparently the French have it right about "salad last."

HOW IT WORKS

I know it's not much fun to see a delicious steak and think "amino acids," but that's what your body does. If you took apart any protein, you would find an array of amino acids.

As I explained in the introduction to this chapter, all living things need amino acids to make protein. The human body needs twenty different amino acids. Nine of these are impossible for our body to make on its own. We get them through the proteins we eat. The proteins that have all nine are considered complete proteins, and we want to consume these proteins, because the body uses them to maintain a healthy immune system, produce certain hormones, and maintain and increase muscle mass, among other functions.[112] You might be amazed to learn that most of

[112] Guoyao Wu, "Amino Acids: Metabolism, Functions, and Nutrition," Amino Acids 37, no. 1 (2009): 1–17, https://doi.org/10.1007/s00726-009-0269-0; National Research Council (US) Subcommittee on the Tenth Edition of the Recommended Dietary Allowances, Recommended Dietary Allowances, 10th ed. (National Academies Press [US], 1989), chap. 6, https://www.ncbi.nlm.nih.gov/books/NBK234922/; Richard B. Kreider and Bill Campbell, "Protein for Exercise and Recovery," Physician and Sportsmedicine 37, no. 2 (2009): 13–21, https://doi.org/10.3810/psm.2009.06.1705; David D. Church et al., "Essential Amino Acids and Protein Synthesis: Insights into Maximizing the Muscle and Whole-Body Response to Feeding," Nutrients 12, no. 12 (2020): 3717, https://doi.org/10.3390/nu12123717.

your organs, not just your muscles, are made of proteins that are remade every few weeks to months. And it's these nine essential amino acids that allow for this to occur. Your spleen, your heart, your liver, your muscles—the protein we consume allows them to keep working as they should.

Your body is smart. While you might think you eat for pleasure, it's actually how your body keeps you seeking the food it needs to maintain these processes—that is, seeking those nine essential amino acids.

Whole grains, vegetables, seeds, and legumes do not contain all nine amino acids, but it is entirely possible to obtain all nine by consuming a variety of these plant-based proteins. Whether you eat meat or not, you are foraging for amino acids as you choose your proteins each day. That's why we feel so satisfied after we eat complete protein; it's because the feeling of satiety is strongly linked to obtaining enough essential amino acids.[113]

And there is one particular amino acid we prioritize in that foraging, even if we are not consciously aware of it. An expanding body of evidence indicates that leucine, one of those nine essential amino acids, is especially impactful.[114]

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[113] M. Veldhorst et al., "Protein-Induced Satiety: Effects and Mechanisms of Different Proteins," Physiology & Behavior 94, no. 2 (2008): 300–307, https://doi.org/10.1016/j.physbeh.2008.01.003.

[114] Shahab Ur Rehman et al., "Research Progress in the Role and Mechanism of Leucine in Regulating Animal Growth and Development," Frontiers in

Rising leucine levels in a cell launch a signaling pathway associated with the protein mTOR (I mentioned it in the introduction to this chapter), which in turn triggers the body to make muscle.[115] This amino acid also has several other known effects. Research has found links to improved immune function, for example,[116] as well as increased muscle mass among older individuals.[117]

As discussed earlier, mTOR has been linked with aging and cancer. One mutant strain of mice lived three times longer than usual after treatment with the drug that lowers mTOR, called rapamycin.[118] But again, those results don't appear to translate to humans, and in fact they have translated to far less dramatic results in normal mice. There is currently no evidence that stimulating mTOR has a life-shortening effect on humans. Rather, eating complete protein foods, and especially those rich in leucine such as lean meat, fish, and eggs, should be the cornerstone of your diet.

Physiology 14 (November 2023): 1252089, https://doi.org/10.3389/fphys.2023.1252089.

[115] Micah J. Drummond and Blake B. Rasmussen, "Leucine-Enriched Nutrients and the Regulation of Mammalian Target of Rapamycin Signalling and Human Skeletal Muscle Protein Synthesis," Current Opinion in Clinical Nutrition and Metabolic Care 11, no. 3 (2008): 222–26, https://doi.org/10.1097/MCO.0b013e3282fa17fb.

[116] Elitsa A. Ananieva et al., "Leucine Metabolism in T Cell Activation: mTOR Signaling and Beyond," Advances in Nutrition 7, no. 4 (2016): 798S–805S, https://doi.org/10.3945/an.115.011221.

[117] Rehman et al., "Research Progress in the Role."

[118] Mikhail V. Blagosklonny, "Rapamycin for Longevity: Opinion Article," Aging 11, no. 19 (2019): 8048–67, https://doi.org/10.18632/aging.102355.

And not to belabor the point, but once again, we are less likely to overeat when we consume sufficient protein because it leaves us feeling full and satisfied.[119]

COMMON QUESTIONS

Eating 1 gram of protein per pound of body weight leads me to gain weight and/or I don't feel good. What should I do

Most people who strive to eat 1 gram of protein per pound of body weight do feel better than they did before adopting that habit. From time to time, though, someone will tell me they just don't feel "right" eating so much protein, and that they feel that they must force themselves to do so, even if their overall caloric intake is in proper range for them. We don't have the exact knowledge to know why that might be. I notice my protein hunger follows the intensity and amount of exercise I do, and while that makes sense based on how protein synthesis works, we don't have any formal studies on training and protein hunger, so take my experience as what it is: anecdotal.

For those of you who struggle to eat 1 gram of quality protein per pound of body weight, there are a few things to consider: Make sure you're eating high-protein foods

[119] Douglas Paddon-Jones et al., "Protein, Weight Management, and Satiety," American Journal of Clinical Nutrition 87, no. 5 (2008): 1558S–61S, https://doi.org/10.1093/ajcn/87.5.1558S.

without excess fat or preservatives. If that does not eliminate the discomfort, then try replacing about a fourth of your protein calories with clean starches, like rice, and with fibrous foods, like salad, broccoli, or apples, while keeping your overall calories approximately the same. Starches are calorie dense, so it won't take much to offset the slightly lower protein intake. Then, make adjustments until you arrive at a way of eating that is sustainable, and makes you feel good.

CICO—calories in, calories out—is a universal law of nutrition. Well, that and we all do need adequate protein and essential fatty acids to survive (which is what this protocol is all about). Other than that, figure out what works for you. And what works best for you may even change over time. I have long made it a point to eat enough protein daily, but I need to eat more vegetables now than I did when I was younger to feel my best.



NUTRITION PROTOCOL

Choose Fibrous Vegetables and (Some) Fruit as Your Major Sources of Carbohydrates

WHAT TO DO

- Vegetables should form the second major brick in the foundation of your diet and fruit, the third. If protein is

the cornerstone, these foods build on it by helping to ensure enough vitamin, mineral, and other micronutrient intake.

- These foods are also essential for our fiber intake. Men should aim for at least 35 grams per day, and women at least 25 grams per day.

HOW IT WORKS

Many readers know already why vegetables and fruits are so important: They contain vitamins, minerals, and other micronutrients like flavonoids, all of which our bodies need to function optimally.[120] Once you understand how nutrient-rich vegetables and fruits are, you won't think about them the same way again. Eating vegetables isn't a chore; it's a gift.

The fiber in vegetables and fruits is also important.[121] A bit about fiber: It provides support structure for the plants that produce it, be that in the roots, the stems, or the leaves. And it's the part we can't digest when we consume them. Fiber can be insoluble (will not dissolve in water), soluble (will dissolve in water), or prebiotic (feeds bacteria).

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[120] Joanne L. Slavin and Beate Lloyd, "Health Benefits of Fruits and Vegetables," Advances in Nutrition 3, no. 4 (2012): 506–16, https://doi.org/10.3945/an.112.002154.

[121] Corina-Bianca Ioniță-Mîndrican et al., "Therapeutic Benefits and Dietary Restrictions of Fiber Intake: A State of the Art Review," Nutrients 14, no. 13 (2022): 2641, https://doi.org/10.3390/nu14132641.

We need all three. Insoluble fiber, present in root vegetables and fruits with edible seeds, helps with stool formation. Soluble fiber, present in foods such as avocados, carrots, and Brussels sprouts, collects waste products as it travels through the body, helping to eliminate them. Prebiotic fiber, present in apples, asparagus, onions, and garlic, provides sustenance for the beneficial bacteria in our gut microbiome, which promote intestinal health and also healthy brain functioning via the gut-brain axis.[122] (The foods noted above are just a few examples from each category.)

And they do a lot more. Studies also indicate that dietary fiber reduces the risk of metabolic syndrome, a cluster of symptoms that increase the risk of heart disease and

[122] Mary Ellen Sanders et al., "Probiotics and Prebiotics in Intestinal Health and Disease: From Biology to the Clinic," Nature Reviews Gastroenterology & Hepatology 16 (July 2019): 605–16, https://doi.org/10.1038/s41575-019-0173-3; Boushra Dalile et al., "The Role of Short-Chain Fatty Acids in Microbiota–Gut–Brain Communication," Nature Reviews Gastroenterology & Hepatology 16, no. 8 (August 2019): 461–78, https://doi.org/10.1038/s41575-019-0157-3; Elizabeth Schneider et al., "Feeding Gut Microbes to Nourish the Brain: Unravelling the Diet–Microbiota–Gut–Brain Axis," Nature Metabolism 6 (2024): 1454–78, https://doi.org/10.1038/s42255-024-01108-6; María Aburto and John F. Cryan, "Gastrointestinal and Brain Barriers: Unlocking Gates of Communication Across the Microbiota–Gut–Brain Axis," Nature Reviews Gastroenterology & Hepatology 21, no. 4 (April 2024): 222–47, https://doi.org/10.1038/s41575-023-00890-0; Robert Hutkins et al., "Classifying Compounds as Prebiotics—Scientific Perspectives and Recommendations," Nature Reviews Gastroenterology & Hepatology 22 (2025): 54–70, https://doi.org/10.1038/s41575-024-00981-6; Yolanda Sanz et al., "The Gut Microbiome Connects Nutrition and Human Health," Nature Reviews Gastroenterology & Hepatology 22 (2025): 534–55, https://doi.org/10.1038/s41575-025-01077-5.

diabetes. The heart-protecting benefits of fiber are widely known. A study that followed 9,776 US men and women for an average of nineteen years found that those who ate the least dietary fiber had a significantly higher risk of coronary heart disease compared to those who ate the most dietary fiber.[123] A review of eleven meta-analyses, which each included several studies, confirmed the link between high fiber consumption and a reduced risk of several cancers, including ovarian, prostate, colon, breast, and pancreatic.[124]

Consuming fiber can also leave us feeling full, simply from the physical bulk of the food in our gut and the slowed gastric emptying that results.[125] It also leaves us less fatigued. That's because consuming adequate fiber helps adjust what's called the postprandial glucose response, the entry of sugar (and fat) into the bloodstream, which can otherwise cause "brain fog" and tiredness. Fiber also blunts post-meal inflammation that often follows eating a dish high in fat and carbs. That inflammation, in which levels of some

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123 Lydia A. Bazzano et al., "Dietary Fiber Intake and Reduced Risk of Coronary Heart Disease in US Men and Women: The National Health and Nutrition Examination Survey I Epidemiologic Follow-Up Study," Archives of Internal Medicine 163, no. 16 (2003): 1897–1904, https://doi.org/10.1001/archinte.163.16.1897.

124 Jun Hu et al., "Use of Dietary Fibers in Reducing the Risk of Several Cancer Types: An Umbrella Review," Nutrients 15, no. 11 (2023): 2545, https://doi.org/10.3390/nu15112545.

125 Eliana Bistriche Giuntini et al., "The Effects of Soluble Dietary Fibers on Glycemic Response: An Overview and Futures Perspectives," Foods 11, no. 23 (December 2022): 3934, https://doi.org/10.3390/foods11233934.

proteins and hormones rise in the body, can contribute to the risk of heart disease if it occurs with any regularity.

How much fiber should we get? Men should get at least 35 grams per day and women 25 grams per day. Very few people actually do that. In fact, an estimated 95 percent of Americans do not consume recommended amounts of fiber.[126]

Some people fear adding more fiber to their diet because it can cause gas and/or bloating. It's true that a sudden increase in daily fiber intake may temporarily result in both. But once you're eating enough fiber on a regular basis, this usually becomes a non-issue. In one study on the link between bloating, sodium, and a high-fiber diet, the researchers found that while this diet did increase bloating to some extent, an accompanying decrease in sodium lessened that side effect, pointing to the idea that it's often what we add to vegetables that becomes the problem, not the vegetables themselves.[127]

And at risk of sounding like your mother, vegetables require more chewing. You need to slow down and chew them well

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[126] Diane Quagliani and Patricia Felt-Gunderson, "Closing America's Fiber Intake Gap," American Journal of Lifestyle Medicine 11, no. 1 (July 2016): 80–85, https://doi.org/10.1177/1559827615588079.

[127] Allison W. Peng et al., "Effects of the DASH Diet and Sodium Intake on Bloating: Results from the DASH-Sodium Trial," American Journal of Gastroenterology 114, no. 7 (2019): 1109–15, https://doi.org/10.14309/ajg.0000000000000283.

for them to digest properly, and for you to reap their full benefits.

A few notes on fruit: Some people find that eating fruit at the end of a meal causes some digestive discomfort. Others have no such issue. If you experience this problem, try eating fruit alone, before a meal, or as a snack rather than to cap off a full meal. It may help.

Beyond fiber, these foods contain, as I mentioned, essential vitamins, minerals, and other micronutrients. The words vitamins and minerals are thrown around a lot (I did it above). But knowing which ones tend to be lacking in most people's diets, and knowing how to get more of these from food, can make a real difference to our health. Magnesium is largely an under-discussed, beneficial component of many foods. This mineral, the eighth most abundant element on earth,[128] is used by the body for energy production, bone development, making DNA, muscle contraction, nerve impulses, and even for regulating our heartbeat.[129] In fact, it's involved in more than three hundred systems in our body and thousands of crucial biological processes. Yet most people don't get enough of it. It's easy to rectify, and I'm

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128 "Magnesium Compounds Statistics and Information," National Minerals Information Center, n.d., https://www.usgs.gov/centers/national-minerals-information-center/magnesium-compounds-statistics-and-information.

129 "Magnesium: Fact Sheet for Health Professionals," National Institutes of Health Office of Dietary Supplements, updated January 6, 2026, https://ods.od.nih.gov/factsheets/Magnesium-HealthProfessional/.

sure you can guess how: by eating enough fruits and especially enough vegetables.

Studies of the so-called Paleo diet, which focuses on fruits, vegetables, lean meat, fish, eggs, and nuts (in other words, foods that likely reflect the diet of a hunter-gatherer living in the Paleolithic era) indicate that we once had a magnesium intake of about 600 milligrams per day, leading researchers to theorize that human metabolism is adjusted to this high level.[130] Health authorities currently recommend 420 milligrams of magnesium daily for adult males and 320 milligrams daily for adult females. Available data indicate that most people consume less than that already low recommended amount. For Americans, the average is about 323 milligrams per day for males and 228 milligrams per day for females. This level is due, in part, to not eating enough foods containing magnesium, but likely also because many foods that would be high in magnesium are grown in soil depleted of this mineral due to modern farming methods.[131] Major or persistent magnesium deficiency can

[130] James J. DiNicolantonio et al., "Subclinical Magnesium Deficiency: A Principal Driver of Cardiovascular Disease and a Public Health Crisis," Open Heart 5, no. 1 (January 2018): e000668, https://doi.org/10.1136/openhrt-2017-000668; Erratum, DiNicolantonio et al., "Subclinical Magnesium Deficiency," Open Heart 5, no. 1 (April 2018): e000668corr1, https://doi.org/10.1136/openhrt-2017-000668corr1.

[131] Ahmad Hassan Chaudhry et al., "Current Understandings on Magnesium Deficiency and Future Outlooks for Sustainable Agriculture," International Journal of Molecular Sciences 22, no. 4 (2021): 1819, https://doi.org/10.3390/ijms22041819.

cause heart problems, neurological problems, and also dangerous reductions in calcium and potassium.[132] And we don't know how moderate deficiencies over time may more subtly diminish our health and vitality.

Vegetables and some fish, such as salmon and mackerel, are excellent sources of magnesium. Avocados (technically a fruit) and certain dairy products, such as milk and cheese, also contain magnesium. Among the foods with the highest amount of magnesium are pumpkin seeds, almonds, black beans, and soybeans. And I know it can be hard to feel excited about kale, spinach, and chard, but a few minutes spent enlightening yourself about magnesium will likely change that. Dark, leafy greens are an excellent source of it and very low in calories; one cup of cooked spinach contains more than a third of the recommended daily intake and fewer than 50 calories.

And there are more vitamins and minerals that vegetables offer. Potassium (beets, tomatoes, butternut squash) is integral to muscle contraction and nerve impulses.[133] Calcium (broccoli, kale, edamame, leafy greens) is crucial

[132] R. Swaminathan, "Magnesium Metabolism and Its Disorders," Clinical Biochemist Reviews 24, no. 2 (2003): 47–66, https://pubmed.ncbi.nlm.nih.gov/18568054/.

[133] Moushumi Sur and Shamim S. Mohiuddin, "Potassium," StatPearls, last updated October 5, 2024, https://www.ncbi.nlm.nih.gov/books/NBK539791/.

for our bones and our teeth.[134] Flavonoids, a category that includes a wide number of compounds present in the plant world, can reduce inflammation; the risk of diabetes, heart disease, and breast cancer; antibiotic resistance; the formation of free radicals (atoms that can damage cells and can contribute to cancer formation); oxidative stress, which can contribute to various diseases and conditions including neurodegenerative disorders; and the risk of viral, bacterial, and fungal infections.[135]

We could continue talking about the benefits of high-quality protein, vegetables, and fruits until the cows (and cauliflower) come home. In fact, cruciferous vegetables such as cauliflower and broccoli offer numerous additional benefits such as sulforaphane, which can help scavenge and detoxify endocrine disruptors and generally supports healthy reproductive hormones in women and men,[136] and on and on. I hope that all we've covered in this protocol has inspired you to fill half your lunch or dinner plate with vegetables.

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[134] Elaine Yu and Sandeep Sharma, "Physiology, Calcium," StatPearls, last updated August 14, 2023, https://www.ncbi.nlm.nih.gov/books/NBK482128/.

[135] A. N. Panche et al., "Flavonoids: An Overview," Journal of Nutritional Science 5 (December 29, 2016): e47, https://doi.org/10.1017/jns.2016.41; Panche et al., "Flavonoids: An Overview—CORRIGENDUM in Figure 1," Journal of Nutritional Science 14 (January 29, 2025): e11, https://doi.org/10.1017/jns.2024.73.

[136] Jed W. Fahey and Mirran Raphaely, "The Impact of Sulforaphane on Sex-Specific Conditions and Hormone Balance: A Comprehensive Review," Applied Sciences 15, no. 2 (January 2025): 522, https://doi.org/10.3390/app15020522.

Perhaps even add some spinach and tomatoes (though, again, technically a fruit) to your breakfast too.

Though one last note on fruit: Most people, unless they are very lean (less than 10 percent body fat for men and less than 15 percent for women) should probably not be eating more than three servings of fruit per day, especially if they are trying to lose body fat. Too much fruit, despite its fiber and water content, can stimulate hunger. Don't get me wrong, fruit is amazing, and is always a part of a healthy diet. Enjoy it, but don't overdo it.

COMMON QUESTIONS

Should I take magnesium supplements

I suggest turning to magnesium supplements such as magnesium bisglycinate or L-threonate before bed only as an if-needed sleep aid, as described in Sleep Protocol 9, and then only if you've tried all the other protocols in chapter 1. For the benefit described here, stick primarily with foods that contain magnesium as a source.

That being said, magnesium supplements can help us get our daily recommended amount of this mineral. Magnesium L-threonate and bisglycinate seem to avoid the issue of gastrointestinal distress that some people experience with this supplement. By contrast, magnesium citrate can cause gastric distress and can sometimes even act as a laxative.

For adults, a 135-milligram magnesium L-threonate or bisglycinate supplement daily before sleep may also help ensure you are obtaining the recommended daily amount. Or you can take it with dinner. Be sure to consult with a physician before adding any supplement into your routine.



NUTRITION PROTOCOL

Consume Healthy Fats as Part of Most Meals and Snacks

WHAT TO DO

- Most of your fat intake should come from unsaturated fats. These are present in olives, avocados, olive oil, and avocado oil, among other foods. Cook with olive oil, and use it for your salad dressing. If you need an oil with a higher smoke point for certain recipes, avocado oil is a good choice.
- Consume omega-3 polyunsaturated fats. These are present in fatty ocean fish, like salmon with the skin, and in various fish oil and krill supplements. (More on the benefits of omega-3 in Nutrition Protocol 7 of this chapter.)

HOW IT WORKS

Perhaps no category of food is as feared, misunderstood, and maligned as fat. Part of the problem is terminology: The fact

that we call dietary lipids fats brings to mind body fat. The two are not the same but they are easy to conflate.

I want you to create a healthy relationship with fat. The purpose of this protocol is to do that, based on science.

We all need some fat in our diet. The question is not if, but which and how much. Which? It turns out there are quite a few to choose from. And how much? As you learned earlier in this chapter, it depends on the other macronutrients you are eating, or avoiding, according to your overall diet (keto, clean omnivore, vegan, and so on).

And yet we have become wary of this macronutrient. This fear is based less on science than on history. Concerns about a link between fat and heart disease date back to the 1940s and grew in the ensuing decades.[137] In 1990, the USDA Dietary Guidelines recommended that Americans limit fat to 30 percent or less of their total calories and saturated fat to just 10 percent of their total calories, a recommendation that led to a proliferation of low-fat and nonfat products.[138] It soon became clear that these foods, often ultra-processed, were much worse for us than the fat we were now avoiding. These guidelines scared many people away from highly

[137] Ann F. La Berge, "How the Ideology of Low Fat Conquered America," Journal of the History of Medicine and Allied Sciences 63, no. 2 (2008): 139–77, https://doi.org/10.1093/jhmas/jrn001.

[138] Dariush Mozaffarian and David S. Ludwig, "The 2015 US Dietary Guidelines: Lifting the Ban on Total Dietary Fat," JAMA 313, no. 24 (2015): 2421–22, https://doi.org/10.1001/jama.2015.5941.

nutritious, high-quality protein foods that also contain some fat, like red meat and eggs. Moreover, we were then living without two crucial benefits that consuming fat provides: the way it slows the rise of glucose in the bloodstream and the way it increases satiety between meals. And Americans, on average, gained weight and suffered from rising rates of heart disease in the following years.

More recently, the pendulum swung the other way with rising interest in ketogenic diets, which involve eating mostly fat, some protein, and extremely few carbohydrates.[139] Interestingly, this diet first originated as a treatment for some forms of epilepsy in the 1920s. It's also been studied as a treatment for schizophrenia[140] and bipolar disorder.[141] As a daily routine, the ketogenic diet is hard to stick with. But this and similar trends helped spread the awareness that many fats, such as monounsaturated and polyunsaturated fats, are good for us.

How can we move forward? Again, by looking at the science. Though a comprehensive review of all types of fat is beyond the scope of this book, a few things are helpful to

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[139] James W. Wheless, "History of the Ketogenic Diet," Epilepsia 49, no. S8 (2008): 3–5, https://doi.org/10.1111/j.1528-1167.2008.01821.x.

[140] Adam Włodarczyk et al., "Ketogenic Diet for Schizophrenia: Nutritional Approach to Antipsychotic Treatment," Medical Hypotheses 118 (September 2018): 74–77, https://doi.org/10.1016/j.mehy.2018.06.022.

[141] Nicole Needham et al., "Pilot Study of a Ketogenic Diet in Bipolar Disorder," BJPsych Open 9, no. 6 (2023): e176, https://doi.org/10.1192/bjo.2023.568.

this discussion. Above we established that fats are made from fatty acids. The body breaks down the fat we eat into these component parts, which are then absorbed into the blood. Once inside the body, fatty acids contribute to the structure and function of cell membranes, help regulate metabolism, and influence hormone production, among other roles.[142] Yes, dietary fat can be more readily converted into body fat, and it's higher in calories per gram than carbs or protein—but at the end of the day, if we are burning as much or more of the calories we eat for energy to meet our metabolic needs, then it doesn't matter if the calories we eat come from fat, carbs, or proteins.

We covered the main types of dietary fat in the introduction to this chapter: monounsaturated, polyunsaturated, saturated. There is also trans fat. A bit more detail on all is warranted. The terms saturated and unsaturated, and the reason the former are solid at room temp and the latter are not, have to do with their different chemical bonds inside the chains of fatty acids.

The lack of double bonds in saturated fats makes the molecules stack on each other more easily, which makes the

[142] Philip C. Calder, "Functional Roles of Fatty Acids and Their Effects on Human Health," Journal of Parenteral and Enteral Nutrition 39, no. 1s (2015): 18S–32S, https://doi.org/10.1177/0148607115595980; Rosanna Chianese et al., "Impact of Dietary Fats on Brain Functions," Current Neuropharmacology 16, no. 7 (2018): 1059–85, https://doi.org/10.2174/1570159X15666171017102547.

fat more stable; this is why products like butter stay solid at room temperature. These fats are hard to digest, which can leave us feeling tired. They can also increase certain forms of cholesterol.

Unsaturated fats are kinked in their structure. When you want to consume fat, these are the ones to prioritize. They contain fatty acids, which trigger circuitries involved with many bodily processes mentioned above.[143] Some fatty acids are directly involved in neuroplasticity, which we can loosely define as the continued ability to learn and form new memories and skills.[144] Fatty acids are also integral to regulating appetite. The presence of fat triggers communication along the gut-brain axis, which prompts the brain to release chemicals that deliver a sense of satiety. In addition, fat in our body releases a hormone called leptin, which activates neural circuitries primarily in the hypothalamus that also result in the feeling of fullness.[145] Leptin is also involved in the signaling pathways related to

[143] Chianese et al., "Impact of Dietary Fats."

[144] Richard P. Bazinet and Sophie Layé, "Polyunsaturated Fatty Acids and Their Metabolites in Brain Function and Disease," Nature Reviews Neuroscience 15, no. 12 (2014): 771–85, https://doi.org/10.1038/nrn3820; Tommy Cederholm et al., "ω-3 Fatty Acids in the Prevention of Cognitive Decline in Humans," Advances in Nutrition 4, no. 6 (2013): 672–76, https://doi.org/10.3945/an.113.004556.

[145] Sean Dornbush and Narothama R. Aeddula, "Physiology, Leptin," StatPearls, last updated April 10, 2023, https://www.ncbi.nlm.nih.gov/books/NBK537038/; M. Romon et al., "Leptin Response to Carbohydrate or Fat Meal and Association with Subsequent Satiety and Energy Intake," American Journal of Physiology-Endocrinology and

the menstrual cycle and sperm production.[146] It is one reason why having enough body fat, but not too much, is needed for libido and fertility. It's the signal from the body to the brain that food resources exist in sufficient abundance to potentially bring another member of our species into the world.

The fourth kind of fats, trans fats, are unsaturated, but the type of chemical bond is different from those in the other unsaturated fats. Although trans fats that occur naturally in some foods like meat and some dairy are relatively neutral or even beneficial to our health, when we hear about trans fats it usually refers to industrial-made versions.[147] These types of trans fats negatively impact cardiometabolic health, and were banned from the US food supply in 2018.[148] When you hear about partially hydrogenated vegetable oil, that is an

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Metabolism 277, no. 5 (1999): E855–61, https://doi.org/10.1152/ajpendo.1999.277.5.E855.

[146] A. M. Salem, "Variation of Leptin During Menstrual Cycle and Its Relation to the Hypothalamic-Pituitary-Gonadal (HPG) Axis: A Systematic Review," International Journal of Women's Health 13 (May 2021): 445–58, https://doi.org/10.2147/IJWH.S309299; Pallav Sengupta et al., "Leptin and Male Reproduction," Asian Pacific Journal of Reproduction 8, no. 5 (2019): 220–26, https://doi.org/10.4103/2305-0500.268143.

[147] Mona A. Hegazy et al., "Hidden Industrial Trans-Fatty Acids: Mechanistic Insights into Dyslipidemia, Cardiovascular Disease, and Metabolic Dysfunction-Associated Steatotic Liver Disease," International Journal of Molecular Sciences 26, no. 23 (December 2025): 11715, https://doi.org/10.3390/ijms262311715.

[148] Angela Amico et al., "The Demise of Artificial Trans Fat: A History of a Public Health Achievement," Milbank Quarterly 99, no. 3 (August 2021): 746–70, https://doi.org/10.1111/1468-0009.12515.

industrial trans fat. Prior to 2018, many commercial products, such as margarine, processed meat, microwave popcorn, and packaged baked goods, made in the US contained trans fat, but no more. Health authorities recommend restricting industrial trans fats to 1 percent of total calories consumed—20 calories of a 2,000-calories-per-day diet, which is less than even one tablespoon. If you travel or live outside the US you'll need to see if industrial trans fats are still used in foods there.

As I note in this protocol, ideally choose monounsaturated fats. Also include some amount of saturated fat.[149] Although some studies have found that switching from saturated fats to polyunsaturated fats reduces the risk of heart disease somewhat,[150] other studies have not confirmed this connection.[151] And a common challenge that can arise when eliminating saturated fat from the diet is that it can lead to an

[149] Ruth Sander, "Saturated Fat May Not Be as Bad for Us After All," Nursing Older People 28, no. 5 (2016): 13, https://doi.org/10.7748/nop.28.5.13.s16.

[150] Dariush Mozaffarian et al., "Effects on Coronary Heart Disease of Increasing Polyunsaturated Fat in Place of Saturated Fat: A Systematic Review and Meta-Analysis of Randomized Controlled Trials," PLOS Medicine 7, no. 3 (2010): e1000252, https://doi.org/10.1371/journal.pmed.1000252.

[151] Zoë Harcombe et al., "Evidence from Prospective Cohort Studies Does Not Support Current Dietary Fat Guidelines: A Systematic Review and Meta-Analysis," British Journal of Sports Medicine 51, no. 24 (2017): 1743–49, https://doi.org/10.1136/bjsports-2016-096550; Christopher E. Ramsden et al., "Re-Evaluation of the Traditional Diet-Heart Hypothesis: Analysis of Recovered Data from Minnesota Coronary Experiment (1968–73)," BMJ 353 (April 12, 2016): i1246, https://doi.org/10.1136/bmj.i1246; Erratum, Ramsden et al., "Re-Evaluation of the Traditional Diet-Heart Hypothesis," BMJ 385 (June 28, 2024): q1450, https://doi.org/10.1136/bmj.q1450.

increased consumption of ultra-processed carbohydrates and added sugar (we reach for a processed snack instead of, say, some cheese or full-fat yogurt), which is also detrimental to health.[152] This often happens when someone decides to eliminate meat, and ends up eating more packaged foods to reduce the effort required to prepare high-protein meals. Ultra-processed foods often contain a lot of calories and little to no fiber. I believe it is better to have a little bit of satisfying saturated fat than to replace it with foods that do not trigger a satiety response from the brain, that the body cannot digest effectively, or that lead to overconsumption.

If you're at this part of the protocol still wondering exactly how much fat to incorporate into your diet: Aim for 10 percent or less of your diet to consist of saturated fat. Keeping your overall fat intake to about 30 to 35 percent of your diet will help enable you to receive the benefits of this macronutrient without weight gain. Learn to love the right fats and your hormones, connective tissues, heart, and palate will all thank you.

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[152] James J. DiNicolantonio et al., "The Evidence for Saturated Fat and for Sugar Related to Coronary Heart Disease," Progress in Cardiovascular Diseases 58, no. 5 (2016): 464–72, https://doi.org/10.1016/j.pcad.2015.11.006.

COMMON QUESTIONS

Should I consume more fat on certain days or post exercise

No. Assuming you're eating 30 percent or so of your calories from fat, and that fat is a mix of saturated fats from high CP:C ratio proteins, and otherwise unsaturated fats such as avocados, olive oil or avocado oil, and some nuts and seeds, there is no reason to spike or drop your fat intake. If anything, we need to increase our clean starch intake, not our fat intake, on days we exercise. That's because intense exercise depletes us of glycogen, a sugar that the body makes as a way to store energy, which eating clean starch can replace.



NUTRITION PROTOCOL

Consume Low-Sugar Fermented Foods

WHAT TO DO

- Consume two to four servings of low-sugar fermented foods per day. Eating foods like sauerkraut, kimchi, pickles in brine (that require refrigeration), kefir, low-sugar kombucha, miso, and natto supports a healthy gut microbiome, which in turn supports all your bodily functions.

HOW IT WORKS

The human microbiome—the bacteria that colonize our skin, mouth, digestive system, and the rest of our body—has become a focal point of both health research and health trends in recent years. Data from the Whitehall II study,[153] which has been tracking the health of thousands of British civil servants between the ages of thirty-five and fifty-five since 1985, found that eating fermented dairy foods like yogurt and kefir reduced the risk of death,[154] as long as coronary heart disease and diabetes were not present. In one surprising study of 710 people, consumption of fermented foods was linked to less social anxiety among those who were prone to it.[155] That finding was correlation, not causation, and it's impossible to know whether eating something like sauerkraut made people less anxious. But connections like these start to make mechanistic sense as we learn more about the microbiome.

[153] Research from the Whitehall II (The Stress and Health Study) is available online here: Whitehall II, UCL Brain Sciences, https://www.ucl.ac.uk/brain-sciences/psychiatry/research/mental-health-older-people/projects/whitehall-ii.

[154] Sabita S. Soedamah-Muthu et al., "Consumption of Dairy Products and Associations with Incident Diabetes, CHD and Mortality in the Whitehall II Study," British Journal of Nutrition 109, no. 4 (2013): 718–26, https://doi.org/10.1017/S0007114512001845.

[155] Matthew R. Hilimire et al., "Fermented Foods, Neuroticism, and Social Anxiety: An Interaction Model," Psychiatry Research 228, no. 2 (2015): 203–8, https://doi.org/10.1016/j.psychres.2015.04.023.

Recent research suggests that our microbiome starts forming before we are even born.[156] Bacteria we are exposed to during the birth process and in the early months of life continue to build this internal environment, which is then added to by microbes we take in through what we eat, what we drink, the people we come into contact with, the air we breathe, even the pets around us, and the soils we come into contact with. The specific makeup of our microbiome, including how diverse it is, changes according to our diet and our environment.[157] Even exercise affects our microbiome. A review of twenty-eight studies found that moderate-to-high-intensity exercise done for thirty to ninety minutes at least three times per week, for at least two months, "is likely to produce changes in the gut microbiota."[158] The reason for this benefit is not fully understood, but may be connected to the changes our body goes through when we exercise, suggesting that temporarily being pushed out of our comfort zone is good for our gut

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156 R. W. Walker et al., "The Prenatal Gut Microbiome: Are We Colonized with Bacteria in Utero?," Pediatric Obesity 12, no. S1 (2017): 3–17, https://doi.org/10.1111/ijpo.12217.

157 Melissa Lee Phillips, "Gut Reaction: Environmental Effects on the Human Microbiota," Environmental Health Perspectives 117, no. 5 (2009): A198–205, https://doi.org/10.1289/ehp.117-a198; Willem M. de Vos et al., "Gut Microbiome and Health: Mechanistic Insights," Gut 71, no. 5 (2022): 1020–32, https://doi.org/10.1136/gutjnl-2021-326789.

158 Alexander N. Boytar et al., "The Effect of Exercise Prescription on the Human Gut Microbiota and Comparison Between Clinical and Apparently Healthy Populations: A Systematic Review," Nutrients 15, no. 6 (2023): 1534, https://doi.org/10.3390/nu15061534.

diversity.[159] Then again, we update our gut microbiome from all sorts of sources around us. The mere act of lacing up our muddy running shoes helps build our microbiome. I know that sounds crazy but that's how the microbiome works: It's an internal reflection of our interactions with the outside world.

Not all bacteria that enter our bodies take up residence in our gut.[160] Some cannot survive the acidic environment, for example. Other bacteria cannot survive in areas with high oxygen concentration. And antibiotics tend to wipe out both the beneficial and detrimental bacteria from our gut.

And it's important to have that beneficial bacteria present there. Studies have found connections between the content of

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[159] Vincenzo Monda et al., "Exercise Modifies the Gut Microbiota with Positive Health Effects," Oxidative Medicine and Cellular Longevity 2017 (March 2017): 3831972, https://doi.org/10.1155/2017/3831972; Belle Huang et al., "Bidirectional Link Between Exercise and the Gut Microbiota," Exercise and Sport Sciences Reviews 52, no. 4 (October 2024): 132–44, https://doi.org/10.1249/JES.0000000000000343; Leizi Min et al., "Effects of Exercise on Gut Microbiota of Adults: A Systematic Review and Meta-Analysis," Nutrients 16, no. 7 (April 2024): 1070, https://doi.org/10.3390/nu16071070; Yousra Alsinani et al., "Exercise and the Gut Microbiome: From Mechanisms to Clinical Applications," Nutrients 18, no. 10 (May 2026): 1565, https://doi.org/10.3390/nu18101565.

[160] Boytar et al., "Effect of Exercise Prescription"; Muriel Derrien and Johan E. T. van Hylckama Vlieg, "Fate, Activity, and Impact of Ingested Bacteria Within the Human Gut Microbiota," Trends in Microbiology 23, no. 6 (June 2015): 354–66, https://doi.org/10.1016/j.tim.2015.03.002; Hilary P. Browne et al., "Transmission of the Gut Microbiota: Spreading of Health," Nature Reviews Microbiology 15 (2017): 531–43, https://doi.org/10.1038/nrmicro.2017.50.

our gut microbiome and many digestive disorders, such as irritable bowel disease and colorectal cancer.

The gut microbiome also appears to influence mood and mental health through the gut-brain axis (introduced in Nutrition Protocol 1).[161] A 2022 review of twenty-four observational studies and nineteen experimental trials, totaling about four thousand participants, identified differences in the beta diversity—which measures how dissimilar the various bacteria types present are to one another—between individuals who had been diagnosed with major depressive disorder and those who had not.[162] Researchers theorize that disruptions to the gut microbiome alter the release of serotonin and dopamine, setting off a cascade that ultimately ends with the release of cytokines, proteins that trigger the immune system into action and

[161] Jeremy Appleton, "The Gut-Brain Axis: Influence of Microbiota on Mood and Mental Health," Integrative Medicine 17, no. 4 (2018): 28–32, https://pubmed.ncbi.nlm.nih.gov/31043907/; John F. Cryan et al., "The Microbiota-Gut-Brain Axis," Physiological Reviews 99, no. 4 (2019): 1877–2013, https://doi.org/10.1152/physrev.00018.2018; Magnus Simrén et al., "Intestinal Microbiota in Functional Bowel Disorders: A Rome Foundation Report," Gut 62 (2013): 159–76, https://doi.org/10.1136/gutjnl-2012-302167; Alina Janney et al., "Host-Microbiota Maladaptation in Colorectal Cancer," Nature 585 (September 2020): 509–17, https://doi.org/10.1038/s41586-020-2729-3.

[162] Sauliha R. Alli et al., "The Gut Microbiome in Depression and Potential Benefit of Prebiotics, Probiotics and Synbiotics: A Systematic Review of Clinical Trials and Observational Studies," International Journal of Molecular Sciences 23, no. 9 (2022): 4494, https://doi.org/10.3390/ijms23094494.

heighten inflammation.[163] Specific inflammatory cytokines have been linked to depression.[164]

On the positive side, we know that when our gut bacteria are provided with sufficient L-tryptophan (the amino acid precursor to serotonin) and short-chain fatty acids, they can readily help synthesize serotonin and signal for our brain to do the same, via our vagus nerve.

The Human Microbiome Project, the American Gut Project, the British Gut Project, and other ongoing large collaborative studies are continuing to uncover the function of the gut microbiome—both its role in healthy biological processes and its influence on chronic and acute ailments. Studies also seek to figure out which specific species of microbiota are beneficial and which are not. As a general rule, diversity is the most important feature of a healthy gut microbiome.[165] Diet plays a huge role in ensuring that our gut hosts a wide range of healthy bacteria.[166] Fiber has long been a focus in this endeavor to ensure a healthy microbiome (more about

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[163] Yu Du et al., "Crosstalk Between the Microbiota-Gut-Brain Axis and Depression," Heliyon 6, no. 6 (2020): e04097, https://doi.org/10.1016/j.heliyon.2020.e04097.

[164] Rai Khalid Farooq et al., "Role of Inflammatory Cytokines in Depression: Focus on Interleukin-1β," Biomedical Reports 6, no. 1 (2017): 15–20, https://doi.org/10.3892/br.2016.807.

[165] Cryan et al., "Microbiota-Gut-Brain Axis."

[166] Abigail J. Johnson et al., "Daily Sampling Reveals Personalized Diet-Microbiome Associations in Humans," Cell Host & Microbe 25, no. 6 (2019): 789–802.e5, https://doi.org/10.1016/j.chom.2019.05.005.

how to prioritize fiber in Nutrition Protocol 4).[167] But fermented foods appear to play an equally if not more significant role as well. Hence this protocol.

Take the study led by Stanford University biologist Justin Sonnenburg comparing how dietary fiber and fermented foods affected the microbiome of thirty-six participants.[168] Half of the individuals consumed a high-fiber diet for ten weeks, increasing their intake of soluble and insoluble fiber by eating more fruits, vegetables, legumes, grains, nuts, and seeds. They also consumed more vegetable protein and decreased the quantity of animal protein and salt. The other half upped their intake of fermented low-sugar foods like yogurt and kefir and also consumed fermented vegetables (sauerkraut, kimchi) and beverages (kombucha and vegetable brine). The researchers collected blood and stool samples before the ten weeks began, at four weeks into the intervention, and at the end of the following six weeks. The participants then began following a diet of their choice—maintaining their study diet or something else—and samples were collected after another four weeks.

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[167] Edward C. Deehan and Jens Walter, "The Fiber Gap and the Disappearing Gut Microbiome: Implications for Human Nutrition," Trends in Endocrinology & Metabolism 27, no. 5 (2016): 239–42, https://doi.org/10.1016/j.tem.2016.03.001.

[168] Hannah C. Wastyk et al., "Gut-Microbiota-Targeted Diets Modulate Human Immune Status," Cell 184, no. 16 (2021): 4137–53.e14, https://doi.org/10.1016/j.cell.2021.06.019.

Analyses of the blood and stool samples revealed that the group focused on fermented foods experienced a significant decrease in nineteen markers of inflammation, including the cytokines interleukin-6, interleukin-10, and interleukin-12b. The researchers also observed beneficial increases in the diversity of the gut microbiomes, as well as decreases in markers of immune system disruption in the low-sugar fermented food condition. The increase in microbial diversity was notably greater among the fermented foods group compared to the high-fiber group. This diversity lasted through the final four weeks of the study, after participants returned to the diet of their choice, indicating that the change was not due to a temporary passing-through of microbes but rather an "ecosystem remodeling," as the researchers call it in their 2022 report in Cell.

COMMON QUESTIONS

What specific fermented foods should I eat

Any that require refrigeration and are low in sugar, and have no added sugars are good choices. Sauerkraut and kimchi are both excellent choices. However, take note that any sauerkraut, pickles, and such that are stable on the shelf have not been fermented and are not recommended. Kombucha is also great, though it contains small amounts of alcohol and is often high in sugar, and should therefore be consumed in

moderation. Natto and miso, made from fermented soybeans, are also good choices and are widely available.

Many fermented foods can be expensive. Yogurt and apple cider vinegar are lower-cost items and easy to find in your grocery store. Kefir is a yogurt drink that is less common but carried by many markets, and also less expensive than some other fermented products.

It is possible to make fermented foods at home, but the process needs careful attention to avoid contamination with harmful bacteria or anything else that can cause a foodborne illness. Tim Ferriss offered a great protocol for homemade sauerkraut in his book The 4-Hour Chef, and that can also be found online. Pickling vegetables in miso is a relatively easy way to make fermented food at home. If you decide to make your own fermented foods, be sure to follow the recipe exactly, including all advised precautions.

I get an unpleasant histamine response from fermented foods—itchy eyes, tingling in my hands, and brain fog. What should I do

This is not uncommon. I've had this experience too from eating broccoli and onions, and I never knew why. When I hosted Sean Mackey, MD, PhD, and chief of the division of Stanford Pain Medicine, on the Huberman Lab podcast, he told me about his experience with gut pain, one that defied all approaches… until someone suggested he cut out

histamine-containing foods like onions.[169] I can live without onions, but love low-sugar fermented foods and the benefits mentioned above are motivation for me to eat them daily. But they can cause a histamine response. So I have taken the histamine enzyme diamine oxidase (DAO) before these meals. It's inexpensive and it solved the issue for me.



NUTRITION PROTOCOL

Consume Fatty Ocean Fish and/or Consider Omega-3 Supplements

WHAT TO DO

- Make a point to consume some fatty, cold water fish once or twice a week. Consider including salmon with the skin, sardines, mackerel, or herring as part of your weekly diet.
- Take supplements of fish oil or krill oil, or, if you're vegan, algae oil. Aim to consume 1,000 milligrams of eicosapentaenoic acid (EPA) per day. Some people will benefit from consuming as much as 2 to 3 grams of EPA per day, although for most people getting that 1 gram of EPA per day will suffice. Some may even wish to ask their doctor to prescribe generic or name brand Lovaza, which is high in omega-3 fatty acids (at 465 mg per

[169] Andrew Huberman, "Dr. Sean Mackey: Tools to Reduce and Manage Pain," Huberman Lab (podcast audio), January 15, 2024, https://www.hubermanlab.com/episode/dr-sean-mackey-tools-to-reduce-manage-pain.

capsule), and is purified of all metals and other toxins. It is now rather inexpensive to obtain given it is available in generic form. I take it daily.

HOW IT WORKS

Nutrition Protocol 5 included the somewhat bold suggestion that we build a healthier relationship with fat. The omega-3 fatty acids are among the best forms of fat. These compounds form polyunsaturated fats, which are found in some fish, some leafy greens, beans, many plant-based oils and some other foods.[170]

As discussed earlier, fatty acids, the building blocks of fat, help with metabolism, how our bodies respond to hormones, physical performance, and more.[171] The body can turn calories into stored fat. But we can't make omega-3 fatty acids. We have to get them through our diet.

And we should make sure to do that because the evidence that omega-3 fatty acids are good for us—our brains, in particular—is persuasive. Research has found positive ties

[170] Fereidoon Shahidi and Priyatharini Ambigaipalan, "Omega-3 Polyunsaturated Fatty Acids and Their Health Benefits," Annual Review of Food Science and Technology 9 (2018): 345–81, https://doi.org/10.1146/annurev-food-111317-095850.

[171] Mauro G. Di Pasquale, "The Essentials of Essential Fatty Acids," Journal of Dietary Supplements 6, no. 2 (2009): 143–61, https://doi.org/10.1080/19390210902861841.

between omega-3s and memory, cognition,[172] the maintenance of mental health and heart health,[173] improved outcomes after heart disease treatment,[174] and relief from depression.[175]

Let's get specific. Modern interest in omega-3 fatty acids can be traced largely to two Danish researchers who became curious about the diet followed by the Greenland Inuit.[176] The low incidence of heart disease and diabetes led the researchers to wonder whether something in their diet, which contained large amounts of seal and whale fat, was staving off these illnesses. Their 1971 paper comparing plasma lipid and lipoprotein levels, which are associated with coronary

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[172] Ibrahim M. Dighriri et al., "Effects of Omega-3 Polyunsaturated Fatty Acids on Brain Functions: A Systematic Review," Cureus 14, no. 10 (2022): e30091, https://doi.org/10.7759/cureus.30091.

[173] Yuriko Adkins and Darshan S. Kelley, "Mechanisms Underlying the Cardioprotective Effects of Omega-3 Polyunsaturated Fatty Acids," Journal of Nutritional Biochemistry 21, no. 9 (2010): 781–92, https://doi.org/10.1016/j.jnutbio.2009.12.004; Heiner C. Bucher et al., "N-3 Polyunsaturated Fatty Acids in Coronary Heart Disease: A Meta-Analysis of Randomized Controlled Trials," American Journal of Medicine 112, no. 4 (2002): 298–304, https://doi.org/10.1016/s0002-9343(01)01114-7.

[174] Mahin Nomali et al., "Omega-3 Supplementation and Outcomes of Heart Failure: A Systematic Review of Clinical Trials," Medicine 103, no. 3 (2024): e36804, https://doi.org/10.1097/MD.0000000000036804.

[175] Yuhua Liao et al., "Efficacy of Omega-3 PUFAs in Depression: A Meta-Analysis," Translational Psychiatry 9, no. 1 (2019): 190, https://doi.org/10.1038/s41398-019-0515-5; Erratum, Liao et al., "Efficacy of Omega-3 PUFAs," Translational Psychiatry 11, no. 1 (2021): 465, https://doi.org/10.1038/s41398-021-01582-6.

[176] H. O. Bang et al., "Plasma Lipid and Lipoprotein Pattern in Greenlandic West-Coast Eskimos," Lancet 297, no. 7710 (1971): 1143–46, https://doi.org/10.1016/S0140-6736(71)91658-8.

artery disease, among a population of Greenland Inuit and a population of Danish citizens (including a small group of Inuit living in Denmark) led the researchers to speculate about a connection between the high content of unsaturated fat in the Inuit diet and low incidence of coronary artery disease as well as diabetes. The publication launched decades of research about the value of omega-3 fatty acids.

Omega-3 fatty acids have been studied in numerous disease settings, with varying results. Among the more striking findings, aside from the clearly beneficial role in reducing the risk of heart disease, is a link to the reduction of depressive symptoms. In one study, researchers randomized sixty people diagnosed with major depressive disorder to treatment with 1,000 milligrams EPA daily, 20 milligrams of the prescription drug fluoxetine (more commonly known as Prozac), or both for eight weeks.[177] Among the forty-eight patients who completed at least four weeks of study treatment, EPA and fluoxetine were equally effective at treating depression, and the combination yielded better outcomes than either one alone.

[177] Shima Jazayeri et al., "Comparison of Therapeutic Effects of Omega-3 Fatty Acid Eicosapentaenoic Acid and Fluoxetine, Separately and in Combination, in Major Depressive Disorder," Australian & New Zealand Journal of Psychiatry 42, no. 3 (2008): 192–98, https://doi.org/10.1080/00048670701827275.

I mentioned a few of these foods in Nutrition Protocol 5, but it bears repeating here for the particular benefits of omega-3 fatty acids they contain.

COMMON QUESTIONS

Fish oil supplements can be expensive. Are they necessary? Which ones are the best

Indeed they can be expensive and concerns about them turning rancid are real. To get 1 gram or more of EPA per day, make sure the source is fresh and from a reputable, third-party tested supplement company. The liquid fish oil (usually cod, mackerel, anchovies, or some combination) sold in dark bottles to protect from light is most cost-effective. Just be sure to keep the bottle refrigerated after opening or it can go rancid. As I mentioned earlier, these days I take Lovaza, which is a prescription high-dose omega-3 capsule that has been purified of metals and other contaminants. It's available in generic form now so it is inexpensive, but you do need a doctor to prescribe it.

I don't like fish oil or fish. Can algae or krill do the job for omega-3s

All these are decent sources of omega-3 fatty acids. For those who are allergic to fish oil or don't want to take it for ethical reasons, algae oil supplements are vegan and rich with omega-3.

One warning: I know a fair number of people who tolerate fish oil well but for whom krill-based omega-3 gives them a rash that itches. I am one such person. So proceed with caution.

What about just eating a lot of fish instead of taking supplements

Again, it is hard to consume enough fish. High-quality fish can be expensive. We also have to be careful about avoiding mercury, which now exists in many species at levels that are too high for frequent human consumption. King mackerel, swordfish, and bigeye tuna are among the fish with the highest mercury levels. Cold-water fish with the lowest mercury levels and that are excellent sources of omega-3s include inexpensive options, like sardines and herring. These and supplementation are going to be the easiest and most cost-efficient approaches to follow. And maybe some sustainably sourced cod, tilapia, or salmon with the skin on one to two times a week, if you like the taste.